

# Research Essentials

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Sydney Informatics Hub  
Core Research Facilities

**[sydney.edu.au/sydney-informatics-hub](https://sydney.edu.au/sydney-informatics-hub)**



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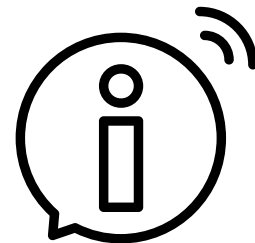
Slides available here

CRICOS 00026A TEQSA PRV12057



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# Acknowledging SIH



- All University of Sydney resources are available to researchers free of charge. The use of the SIH services including the Artemis HPC and associated support and training warrants acknowledgement in any publications, conference proceedings or posters describing work facilitated by these services.
- The continued acknowledgment of the use of SIH facilities ensures the sustainability of our services.

## **Suggested wording for use of workshops and workflows:**

- *“The authors acknowledge the Statistical workshops and workflows provided by the Sydney Informatics Hub, a Core Research Facility of the University of Sydney.”*

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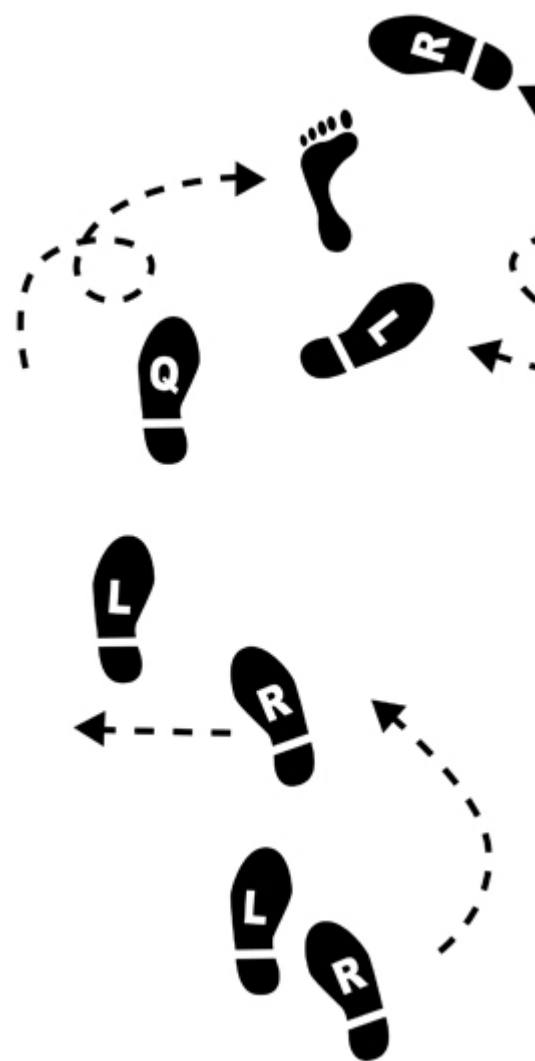
# What is a workflow?

- Every statistical analysis is different, but all follow similar paths. It can be useful to know what these paths are.
- We have developed practical, step-by-step instructions that we call 'workflows', that you can follow and apply to your research.
- We have a general research workflow that you can follow from hypothesis generation to publication.
- And statistical workflows that focus on each major step along the way (e.g. experimental design, power calculation, model building, analysis using linear models/survival/multivariate/survey methods).

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# Statistical workflows

- Our **statistical workflows** can be found within our workshop slides.
- **Statistical workflows** are software agnostic, in that they can be applied using any statistical software.
- To access these statistical workflows and more, visit our [Workshops and Workflows](#) page.



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## Software workflows



- There may also be accompanying **software workflows** that show you how to perform the statistical workflow using particular software packages (e.g. R or SPSS). We won't be going through these in detail during the workshop. If you are having trouble using them, we suggest you attend our monthly [Hacky Hour](#) where SIH staff can help you.
- Our software workflows contain:
  - o R code and comments.
  - o SPSS syntax as well as screenshots of the point and click procedures and written methods.
  - o Screenshots of the point and click procedures and written methods for other bespoke software.

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## During the workshop



- Ask short questions or clarifications during the workshop (either by Zoom chat or verbally). There will be breaks during the workshop for longer questions.



- Slides with this blackboard icon are mainly for your reference, and the material will not be discussed during the workshop.



- Challenge questions will be encountered throughout the workshop.

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# Research Essentials workshop overview

## **8-step general research workflow and other resources**

- Where does this Workshop fit into the research process?
- Where does it fit in with other SIH training and support on offer?

## **Setting up your data for most analyses:**

- Step 3: Collect and store data
- Step 4: Cleaning data

## **Workflow examples for common analyses – brief introduction to:**

- Step 5: Exploratory data analysis
- Step 6: Inferential analysis

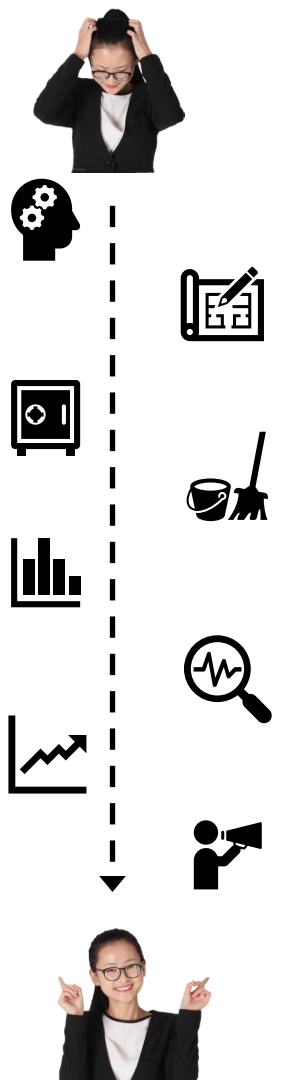
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# 8-step general research workflow

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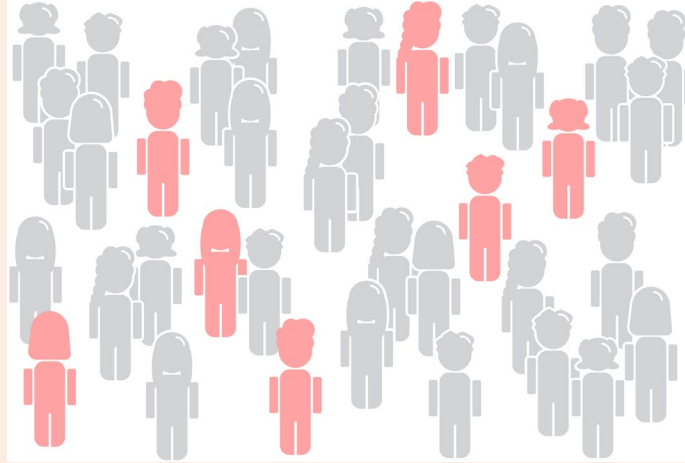
# General research workflow

1. **Hypothesis Generation** (Research/Desktop Review)
2. **Experimental and Analytical Design** (Sampling, power, ethics approval)
3. **Collect/Store Data**
4. **Data cleaning**
5. **Exploratory Data Analysis (EDA)**
6. **Data Analysis aka inferential analysis**
7. **Predictive modelling**
8. **Publication**



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## 6. Statistical Inferential analysis – from sample to population



### **Statistical inference:**

“The theory, methods, and practice of forming judgments about the parameters of a population, usually on the basis of random sampling.”

*Collins English Dictionary*

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## 7. Predictive modelling – Inferential predictive statistics versus machine learning predictive analytics

- **Inferential predictive statistics**



Predicting temperature rise in climate change.

- **Machine learning/predictive analytics**



Very accurately verify fingerprint to unlock a mobile phone.



# Ecosystem of SIH statistical training

| Workflow step                | Other training                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Hypothesis generation     | <a href="#"><u>Research Essentials</u></a>                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2. Study design              | <a href="#"><u>Experimental Design</u></a><br><a href="#"><u>Power and Sample Size</u></a><br><a href="#"><u>Design and Analysis of Surveys 1</u></a> + <a href="#"><u>Design and Analysis of Surveys 2: Advanced Topics</u></a><br><a href="#"><u>Statistical Model Building</u></a>                                                                                                                                                         |
| 3. Collect/store data        | <a href="#"><u>Research Essentials</u></a>                                                                                                                                                                                                                                                                                                                                                                                                    |
| 4. Data cleaning             | <a href="#"><u>Research Essentials</u></a>                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5. Exploratory data analysis | <a href="#"><u>Research Essentials</u></a>                                                                                                                                                                                                                                                                                                                                                                                                    |
| 6. Inferential analysis      | <a href="#"><u>Linear Models 1 to 3</u></a> + <a href="#"><u>Statistical Model Building</u></a><br><a href="#"><u>Introduction to Survival Analysis</u></a><br><a href="#"><u>Meta-Analysis: An Introduction</u></a><br><a href="#"><u>Design and Analysis of Surveys 1</u></a> + <a href="#"><u>Design and Analysis of Surveys 2: Advanced Topics</u></a><br><a href="#"><u>Multivariate Statistical Analysis 1: Dimension Reduction</u></a> |
| 7. Predictive modelling      | [Predictive analytics: <i>Introduction to machine learning in R/Python (SIH Data Science)</i> ]: <a href="#"><u>From 26th August</u></a>                                                                                                                                                                                                                                                                                                      |
| 8. Publication               |                                                                                                                                                                                                                                                                                                                                                                                                                                               |



# SIH training

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Different offerings, in person, online, and hybrid content in a variety of formats from webinars to interactive workshops.



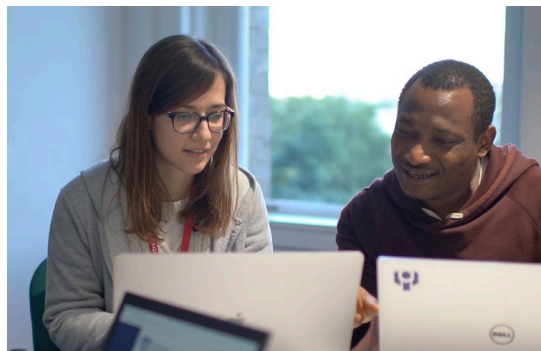
Attendees at all career levels, from undergraduate students to senior professors, and representation from every Faculty and School.



Partnerships with national organisations like Australian BioCommons:  
[biocommons.org.au/training-cooperative](https://biocommons.org.au/training-cooperative).



Find out more on our training calendar:  
[sydney.edu.au/informatics-hub/training](https://sydney.edu.au/informatics-hub/training). Or stay up to date with our newsletter.



| Statistics   | Data Science                | Research Computing         | Bioinformatics                  | Events            |
|--------------|-----------------------------|----------------------------|---------------------------------|-------------------|
| Fundamentals | Machine Learning            | High Performance Computing | 'omics Techniques               | Hacky Hour        |
| Modelling    | Visualisation               | Cloud Computing            | Reproducible Pipelines          | Summer Schools    |
| Specialist   | Natural Language Processing | Containers                 | Data Analytics                  | Coding Challenges |
|              | Geospatial Analysis         | Workflows                  | National Compute Infrastructure |                   |



## Ecosystem of other USyd training

| Workflow step                | Other training                                                                                                                                             |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Hypothesis generation     | Library research support: <a href="#">Literature and systematic review</a>                                                                                 |
| 2. Study design              |                                                                                                                                                            |
| 3. Collect/store data        | RedCap- <a href="#">Various trainings for survey data from introduction to advanced</a><br><a href="#">Research Data Management modules and techniques</a> |
| 4. Data cleaning             |                                                                                                                                                            |
| 5. Exploratory data analysis | <a href="#">SIH: EDA in R</a>                                                                                                                              |
| 6. Inferential analysis      |                                                                                                                                                            |
| 7. Predictive modelling      |                                                                                                                                                            |
| 8. Publication               | Library research support: <a href="#">Publishing</a>                                                                                                       |



# Research data management

- eNotebook
- RedCap
- Research data store
- OneDrive (Office 365)

Supported  
platforms



- Google Drive
- Survey Monkey
- Portable Drives e.g. USB

Do not  
use



[University-supported research data platforms: Features and comparisons](#)

Research data that is managed optimally improves research efficiency and reach, as well as ensuring its integrity and security, and meeting legislative/policy/funding/publishing requirements.

The Research Data Consulting team assists researchers to enhance their research productivity and improve data management practices. They provide:

- Short consultations to integrate digital tools and data management into your research.
- Training and functional support for university supported tools/platforms.

# Research Computing



Research Problems



Artemis PC + RDS

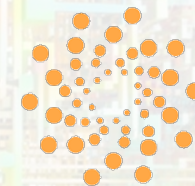
**RONIN**

Institutional

aws



Commercial



NCI  
AUSTRALIA



pawsey

National



Research Insights

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# Hacky Hour

*For researchers who code or analyse data*



- A monthly meetup where anyone from the University – students, staff and university affiliates – can collaborate and get support e.g., swap notes, get help, or learn new techniques in programming and data science.
- Experts & mentors from SIH and across the University will be available to advise and answer questions on coding, data analytics or digital tools.
- Come join us on zoom the **3rd Wednesday of every month, 2 - 3pm!**
- [sydney.edu.au/informatics-hub/hacky-hour](https://sydney.edu.au/informatics-hub/hacky-hour)



# How to engage with us



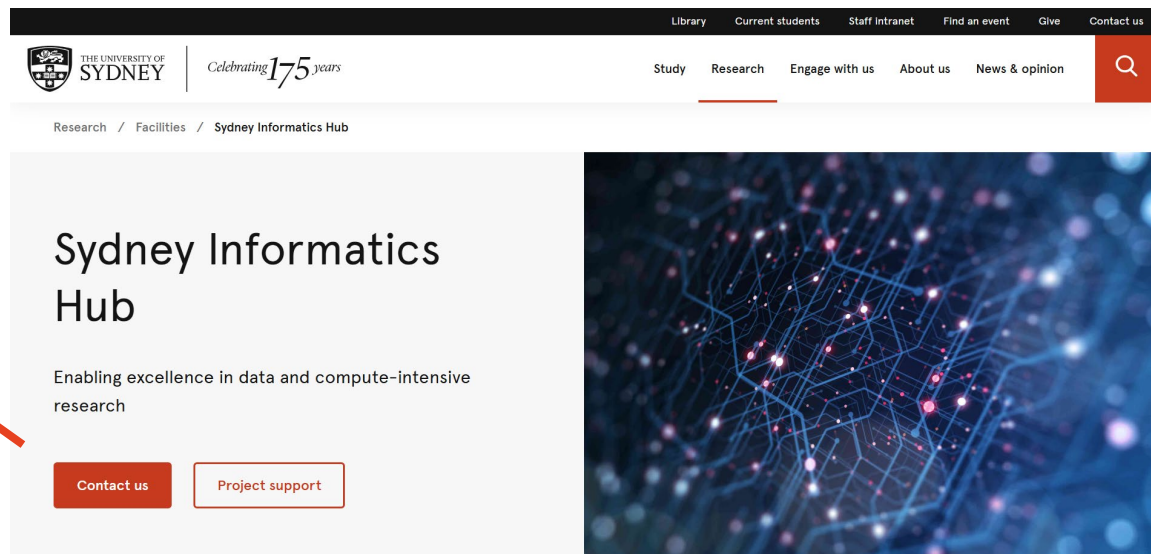
[sydney.edu.au/informatics-hub](https://sydney.edu.au/informatics-hub)



And fill out our request form

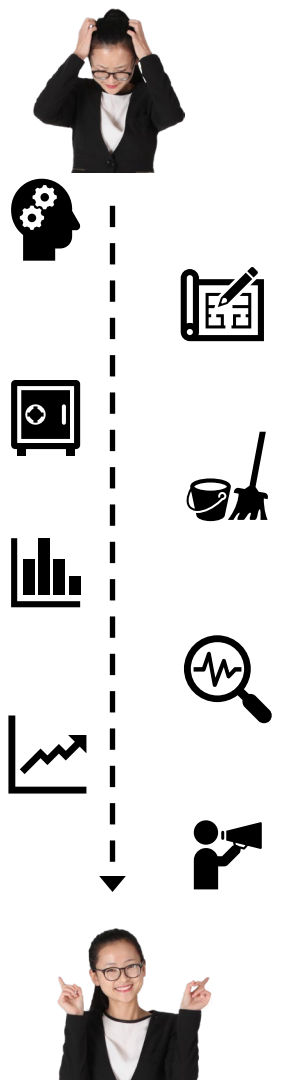


Or send us an email:  
[sih.info@sydney.edu.au](mailto:sih.info@sydney.edu.au)



# General research workflow

1. **Hypothesis Generation** (Research/Desktop Review)
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If statistical programs were cars...

## The first question: Which car will you take?

Excel



Stata



SPSS



R



Python



SAS



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# The first question: Which car will you take?

Getting from step 3 to step 8 will involve using software. Will it be:

**Graphical User Interface (GUI)** – like an automatic car  
Interactive, point and click and  
Menu driven  
Easier to get started



**Command line interface (CLI)** – like a manual car  
Writing code  
Easier to handle complex  
and/or large data sets

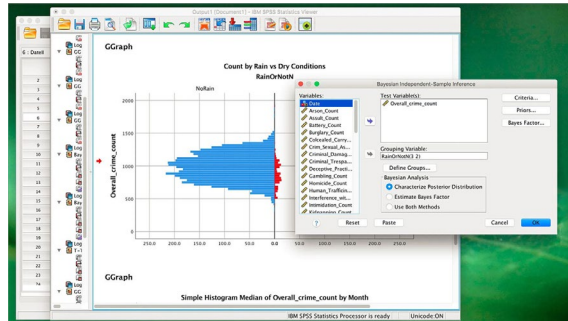
CLI



# Software choice as the first question: GUI versus CLI

- Which software are you more familiar or comfortable with?
- How do you record your analysis for reproducible research?
- By documenting, you should always be able to re-run your analysis from start to finish (and get the same result)!
- If using interactive processing, you should keep track of the commands you run.

## Graphical User Interface (GUI) **SPSS**



## Command line interface (CLI) **R** code

```
R
R> lm(Sepal.Length ~ Species, data = iris)

Call:
lm(formula = Sepal.Length ~ Species, data = iris)

Coefficients:
(Intercept)  Speciesversicolor  Speciesvirginica
          5.006              0.930              1.582

R> |
```



---

## Statistical software used at USyd

- A variety of statistical software options are supported at USyd, and what you use may depend on your lab or faculty or school.
  - The main two we see in our statistical consulting are R (CLI) and SPSS (GUI).
  - We also consult with clients using jamovi, SAS, Stata, Graphpad Prism, Python and Genstat.
- To access these and more:
  - As a student you may need to access via the citrix workspace: [Student IT: Apps](#)
  - As a researcher (with valid staff ID): [What software is available to University staff?](#)

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## 3. Collect/store your data

*A. Research data management*

*B. Organise your data for input into statistical software*

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## A. Research data management

- **Data storage**
- Back up EVERYTHING including original data collection forms or raw data (images, electrical signals, DNA sequences, whatever).
- **Data entry - will you be using manual data entry?**
- Ideally double-data entry followed by comparison.
- Be wary of spreadsheets – especially entering, editing analysing in the same sheet.
- Statistical software generally doesn't allow easy editing once you have entered your data.



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## A. Research data management

- **Have you got a Research data management plan according to University policy?**
  - What is a Research Data Management Plan (RDMP)?
  - What are the university supported tools for data collection and storage?
    - What is an eNotebook?
    - Where can I store my data?
- **Consider appropriate folder/directory structure, file naming and version control for your project, *or at least your part of it.***
  - Good enough practices for scientific computing
  - Best practices for data management and sharing in experimental biomedical research



# A. Research data management

## Guide to storing and managing your projects research data

### University supported and licenced platforms

| Platform/Tool                    | eNotebook           | REDCap                                             | Research Data Store (RDS)                       | OneDrive (Enterprise) | Teams (Enterprise)                 | Highly Protected SharePoint (Enterprise) | Australian Imaging Service (AIS) | Local storage, USB Drive       | Other cloud tools (e.g. Google Drive, Dropbox) |
|----------------------------------|---------------------|----------------------------------------------------|-------------------------------------------------|-----------------------|------------------------------------|------------------------------------------|----------------------------------|--------------------------------|------------------------------------------------|
| function                         | electronic notebook | survey and data capture, including Clinical trials | networked data storage, large files, HPC access | cloud storage         | chat, collaboration, cloud storage | collaboration, cloud storage             | imaging repository and analytics | removable media, local storage | cloud storage                                  |
| suitable for data classification | ●●●                 | ●●●                                                | ●●●                                             | ○●●                   | ○●●                                | ●●●                                      | ●●●                              | ●                              | ●                                              |
| stored in Australia              | ✓                   | ✓                                                  | ✓                                               | ✓                     | ✓                                  | ✓                                        | ✓                                | various                        | ✗                                              |
| external collaborator access     | ✓                   | ✓                                                  | ✓                                               | ✓                     | ✓                                  | ✓                                        | ✓                                | ✗                              | ✗                                              |
| context and commentary supported | ✓                   | ✗                                                  | ✗                                               | ✓                     | ✗                                  | ✗                                        | n/a                              | ✗                              | ✗                                              |
| syncing with local copy          | n/a                 | n/a                                                | n/a                                             | ✓                     | ✓                                  | ✓                                        | n/a                              | ✗                              | ✗                                              |
| available storage                | unlimited           | unlimited                                          | unlimited (default 2TB)                         | 5TB                   | 2TB+                               | 25TB max (default 2TB)                   | unlimited                        | ✗                              | ✗                                              |
| backup and disaster recovery     | ✓                   | ✓                                                  | ✓                                               | ✓                     | ✓                                  | ✓                                        | ✓                                | ✗                              | ✗                                              |
| audit trail/version control      | ✓                   | ✓                                                  | ✓                                               | ✓                     | ✓                                  | ✓                                        | ✓                                | ✗                              | ✗                                              |
| versioning retained              | ✓                   | manual                                             | up to 60 days                                   | 7 years               | 7 years                            | 7 years                                  | ✗                                | ✗                              | ✗                                              |



Version 6.0  
December 2022  
Endorsed by CIO



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|   |                                                        |
|---|--------------------------------------------------------|
| ● | highly protected                                       |
| ○ | highly protected data needs additional file encryption |
| ● | protected                                              |
| ● | public                                                 |

Highly Protected data may require additional encryption depending on some platforms. Protected data may benefit from encryption.

For more information about research data classifications, go to <https://sydney.edu.au/research-data-classifications>

For research data management enquiries, please contact [digital.research@sydney.edu.au](mailto:digital.research@sydney.edu.au)

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# Version control – keeping track of your files

- Use a separate directory for each discrete analysis.
- When processing data and intermediate files save with a new name.
  - Use a good file naming convention to keep track of files.
- Save frequently so if you lose a version, you do not have to redo too much work.
- For collaborative research consider using eNotebook or other version control systems, e.g. Git (free) or GitHub.
- Create a log file in the same directory and use version control (e.g. name sequentially, date/time stamp, for example:
  - “20230208 stats101 workshop v2.0.xlsx” (orders files chronologically).

---

# Version control – keeping track of your files

## Example of a version log file in Excel:

| File name              | Description                                                                                                                                                                                                                                                                               | # Obs | # Vars |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| Mydata_v1_30102022.csv | Original data entry by KS, 1 record per person                                                                                                                                                                                                                                            | 250   | 34     |
| Mydata_v2_01112022.csv | Eligible records only based on study inclusion criteria with new variables created for analysis:<br>BMI calculated from recorded height and weight;<br>babies age processed to be consistently in months instead of days and weeks as well;<br>number of pets categorised (none, 1-2, 3+) | 204   | 37     |

## Data formats – tidy data

- Depending on the design of your experiment/survey you may have a mix of demographic data on each individual, and measurements.
  - You may need multiple tables and a unique ID for each individual to link them, or just have the demographic data repeated when transforming to long format
- Wide and long can become relative terms especially if you have clusters of subjects.
- Tidy data is a consistent way to should have:
  - One variable in each column
  - One observation per row
  - One value per cell

| country     | year | cases | population |
|-------------|------|-------|------------|
| Afghanistan | 1999 | 745   | 19987071   |
| Afghanistan | 2000 | 666   | 20035360   |
| Brazil      | 1999 | 3737  | 17206362   |
| Brazil      | 2000 | 8488  | 17404898   |
| China       | 1999 | 21258 | 1272015272 |
| China       | 2000 | 21366 | 1280058583 |

variables

| country     | year | cases | population |
|-------------|------|-------|------------|
| Afghanistan | 1999 | 745   | 19987071   |
| Afghanistan | 2000 | 666   | 20035360   |
| Brazil      | 1999 | 3737  | 17206362   |
| Brazil      | 2000 | 8488  | 17404898   |
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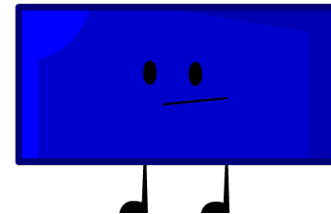
observations

| country     | year | cases | population |
|-------------|------|-------|------------|
| Afghanistan | 1999 | 745   | 19987071   |
| Afghanistan | 2000 | 666   | 20035360   |
| Brazil      | 1999 | 3737  | 17206362   |
| Brazil      | 2000 | 8488  | 17404898   |
| China       | 1999 | 21258 | 1272015272 |
| China       | 2000 | 21366 | 1280058583 |

values

---

## B. Organising a dataset for analysis



- **Most programs read in data in a rectangular format:**
  - E.g. A text file – you can read it in Notepad or any text editor or Excel, csv etc.
  - A header including column names in the first row.
  - Each row thereafter being the data itself (often corresponding to a single unit of interest – e.g. person, animal, plant, plot, farm, machine, business, school, hospital etc).
  - Each column represents one variable.
    - ID variable – identifies the subject.
    - Demographic variable – characteristics of the subject. including their treatment.
    - Measurement variable – some observation on the subject.
  - A delimiter between each column (comma .csv and tab .tsv/.tab/.txt).
  - Do a quick skim of your dataset and look at the corners to see if it is actually rectangle.

---

# Pitfalls when coming from Excel

- Watch out for:
  - Merged cells
  - Cell comments
  - Colour coding
  - Blank rows
  - Data in multiple sheets
  - Particular coding of missing data/blanks/non-applicable
- Deal with the above in Excel before exporting to text. Sometimes these have been added to annotate the data, or make it easier to read. Other times, annotations must be represented in text file.
- A good summary of these pitfalls is provided in [this paper](#).
- Check your data once it is imported into the statistical software.



## B. Data formats – transformations

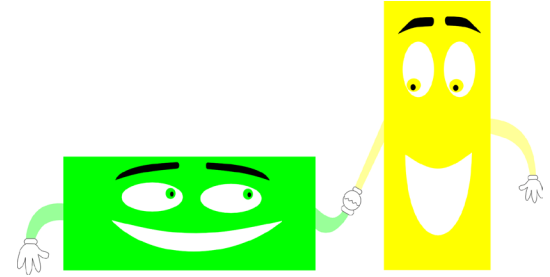
| Patient ID | Time 1 | Time 2 | Time 3 |
|------------|--------|--------|--------|
| 1          | 50     | 55     | 60     |
| 2          | 47     | 49     | 50     |
| ...        | ...    | ...    | ...    |

**Wide/unstacked  
format**



| Patient ID | Time | Body weight |
|------------|------|-------------|
| 1          | 1    | 50          |
| 1          | 2    | 55          |
| 1          | 3    | 60          |
| 2          | 1    | 47          |
| 2          | 2    | 49          |
| 2          | 3    | 50          |
| ...        | ...  | ...         |

**Long/stacked  
format**



---

## B. Organising a dataset for statistical analysis – data coding

- Specify type of variable: ensure your analysis software knows whether a variable is numeric (continuous or discrete), or categorical/factor/string/character (text).
- Label variables, either within the software or by keeping your own record (e.g. Age = Age at interview in years).
- Label variable values/'levels' within categorical variables. While numerical codes are often displayed when you extract survey data, e.g. from your REDCap project, the data may come without label definitions, so keep track of these, e.g. 1 = "Male", 2 = "Female", 3 = "Non-binary".
- Correctly code missing values according to software program: ensure your analysis software knows that the data is missing and not '0' or some other value (e.g. 999, 777, 99, 77).

---

## 4. Data cleaning



# Data cleaning: data wrangling and data dictionary

A simple example of a data dictionary:

|   | A             | C          | D         |
|---|---------------|------------|-----------|
| 1 |               |            |           |
| 2 | Questions     | Categories | Code used |
| 3 | Q1_Age(years) | 20-30      | 1         |
| 4 |               | 31-40      | 2         |
| 5 |               | 41-50      | 3         |
| 6 |               | 51-60      | 4         |
| 7 |               | >60        | 5         |
| 8 | Q2_Gender     | male       | 1         |
| 9 |               | female     | 2         |

- Data cleaning involves examining\* the variables in the dataset and creating new variables for analysis by recoding/processing variables as required.
- Use short but informative variable names; it's a good idea to have a data dictionary.
- Names should keep track of transformations/recoding, e.g.
  - age = original data in years
  - age\_c2 = age categorised into two categories (young vs old)

\* More on how to examine, i.e. describe the distribution of variables later in this workshop.



# Data cleaning: data wrangling and data dictionary

- If you are using R, the janitor package is great for cleaning up variable names.
  - [janitor: Simple Tools for Examining and Cleaning Dirty Data](#)
- Regardless of your software choice, below are some suggested styles for naming your variables:

| Case Style           | Argument (in R)                         | Example     |
|----------------------|-----------------------------------------|-------------|
| Snake case           | "snake"                                 | employee_id |
| Lower camel case     | "lower_camel"                           | employeeld  |
| Upper camel case     | "upper_camel"                           | EmployeeId  |
| Screaming snake case | "screaming_snake"                       | EMPLOYEE_ID |
| Lower case           | "lower"                                 | employeeid  |
| Upper case           | "upper"                                 | EMPLOYEEID  |
| Title case           | "title"                                 | Employee Id |
| Sentence case        | "sentence"                              | Employee id |
| Pascal case          | "pascal" (alias for "upper_camel")      | EmployeeId  |
| Small camel case     | "small_camel" (alias for "lower_camel") | employeeld  |

---

# Keep track of analyses

- **Remember you should be able to repeat analysis from the start, to demonstrate/enable reproducibility.**
- **For statistical programming languages:**
  - Name the program file logically.
  - Use structure, work in blocks or ‘chunks’ of code for different sections, e.g. ‘descriptive analyses’ – do it for all predictors in one go.
  - Log file – same name as program file, different extension – VERY important as record for interactive mode!
  - Use functions to avoid repetition.
  - Use appropriate level of comments, e.g. key steps and results.
  - Consider using an Rmarkdown/Quarto notebook if using R.
- Also covered in [Good enough practices in scientific computing](#).

# Example R Markdown and Quarto files

Files Plots Packages Help Viewer Jobs

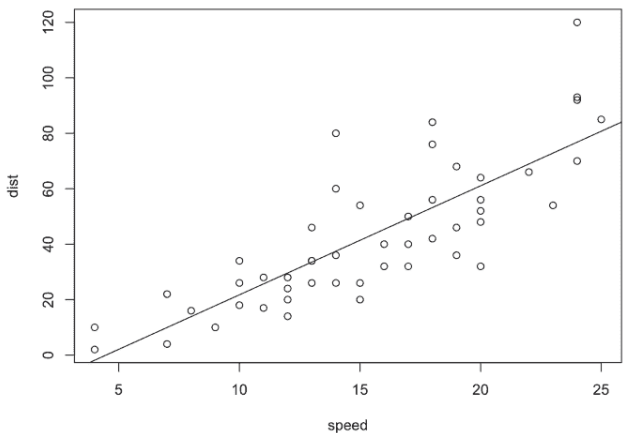
## Hello R Markdown

*Awesome Me*  
2018-02-14

This is a paragraph in an R Markdown document.

Below is a code chunk:

```
fit = lm(dist ~ speed, data = cars)
b = coef(fit)
plot(cars)
abline(fit)
```



The slope of the regression is -17.5790949.

## R for Data Science: Chapter 27 R Markdown

## R for Data Science: Chapter 28 Quarto

markdown.qmd

Render on Save Render

Source Visual / gh Normal Format Insert Table

### Text formatting

*italic* **bold** underline ~~strikeout~~ SMALL CAPS code superscript<sup>2</sup> and subscript<sub>2</sub>

### Headings

#### 1st Level Header

#### 2nd Level Header

#### 3rd Level Header

### Lists

- Bulleted list item 1
- Item 2
  - Item 2a
  - Item 2b

1. Numbered list item 1
2. Item 2. The numbers are incremented automatically in the output.

### Links and images

<http://example.com>

[linked phrase](#)

<http://example.com>



optional caption text

W: 330 h: 77 px 1 Lock ratio

### Tables

| First Header | Second Header |
|--------------|---------------|
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type to search...

- R Code Chunk  
Executable R chunk
- Python Code Chunk  
Executable Python chunk
- DIV...  
Block containing other con...
- Bullet List  
List using bullets for items
- Numbered List  
List using numbers for items
- H1  
Heading 1  
Part heading

Tables 2

Quarto 1

# Data cleaning in action

What are some obvious things about this file that need updating?

|    | A                                                                                  | B          | C         | D      | E            | F | G             | H         | I             | J         | K         | L        | M                    | N                          | O           | P          |
|----|------------------------------------------------------------------------------------|------------|-----------|--------|--------------|---|---------------|-----------|---------------|-----------|-----------|----------|----------------------|----------------------------|-------------|------------|
| 1  | Acute Kidney Injury (AKI) Study                                                    |            |           |        |              |   |               |           |               |           |           |          |                      |                            |             |            |
| 2  | A. Motivated Resident                                                              |            |           |        |              |   |               |           |               |           |           |          |                      |                            |             |            |
| 3  | Research Project                                                                   |            |           |        |              |   |               |           |               |           |           |          |                      |                            |             |            |
| 5  | study_id                                                                           | dob        | Race      | Sex    | Hispanic     |   | Admission GFR | Day 1 gfr | DAY2GFR*      | day 3 Gfr | Day 4 GFR | DAY5 GFR | diagnosis            | sediment                   | HD catheter | AV fistula |
| 6  | 1                                                                                  | 11/11/1967 | White     | Male   | Hispanic     |   | 17            | 12        | 10            | 22        | 34        | 41       | hypotension          | many granular casts        | Yes         | No         |
| 7  | 2                                                                                  | 11/12/1990 | Caucasian | Female | Not Hispanic |   | 21            | 17        | 19 but poor c | 12        | 19        | 28       | sepsis               | muddy brown casts          | Yes         | No         |
| 8  | 3                                                                                  | may 5 1970 | White     | Male   | Hispanic     |   | 32            | 28        | 15            | 11        | 15        | 21       | bleeding out         | many epithelial cell casts | No          | No         |
| 9  | 4                                                                                  | 3-Jun-81   | White     | Female | Not Hispanic |   | 11            | 9         | 6             | 5         | 9         | 18       | norovirus diarrhea   | muddy brown casts          | YES         | YES        |
| 10 | 5                                                                                  | 7/06/1984  | Af-Am     | Male   | Not Hispanic |   | 15            | 12        | 9             | 8         | 11        | 14       | sepsis               | muddy brown casts          | YES         | No         |
| 11 | 6                                                                                  | 9/04/1980  | Black     | Female | H            |   |               |           |               | 26        | 38        | 44       | liver failure        | many epithelial cell casts | No          | No         |
| 12 | 7                                                                                  | 13/01/1985 | White     | Female | Hispanic     |   |               |           |               | 33        | 39        | 43       | pneumonia sepsis     | many granular casts        | No          | No         |
| 13 | 8                                                                                  | 15/11/1968 | Asian     | F      | Hispanic     |   |               |           |               | 38        | 47        | 51       | line sepsis          | many epithelial cell casts | No          | No         |
| 14 | 9                                                                                  | 22/04/1982 | Mixed     | Male   | Not His      |   |               |           |               | 39        | 36        | 41       | C diff diarrhea      | many epithelial cell casts | No          | No         |
| 15 | 10                                                                                 | 12/10/2003 | White     | M      | Not His      |   |               |           |               | 9         | 11        | 14       | NSAID tox            | muddy brown casts          | Yes         | No         |
| 16 | 11                                                                                 | 6/11/1982  | White     | F      | NH           |   |               |           |               | 29        | 36        | 41       | burns                | muddy brown casts          | Yes         | Yes        |
| 17 | 12                                                                                 | 9/07/1979  | Caucasian | M      | Not His      |   |               |           |               | 37        | 44        | 48       | pyelo                | many epithelial cell casts | No          | No         |
| 18 | 13                                                                                 | 5=11-1984  | Black     | F      | NH           |   |               |           |               | 33        | 39        | 44       | urethral obstruction | many epi cell casts        | Yes         | No         |
| 19 | 14                                                                                 | 7/04/2004  | Af-Am     | M      | Hispanic     |   |               |           |               | 41        | 44        | 49       | GI bleed             | many granular casts        | No          | No         |
| 20 | 15                                                                                 | 8/03/1990  | Asian     | M      | NH           |   |               |           |               | 36        | 39        | 43       | sepsis               | rare gran casts            | No          | No         |
| 21 | 16                                                                                 | 6/09/1983  | WHITE     | Male   | Not His      |   |               |           |               | 39        | 47        | 51       | salmonella           | many granular casts        | No          | No         |
| 22 | 17                                                                                 | 11/07/1986 | WHITE     | Female | Not Hispanic |   |               |           |               | 34        | 48        | 52       | sepsis               | rare granular casts        | No          | No         |
| 23 | 18                                                                                 | 9/06/1987  | Black     | Male   | Hispanic     |   |               |           |               | 24        | 39        | 43       | esophageal varices   | many epithelial cell casts | Yes         | No         |
| 24 | 19                                                                                 | 29/03/1979 | Caucasian | Female | Hispanic     |   |               |           |               | 19        | 15        | 18       | dehydration          | many granular casts        | No          | No         |
| 25 | 20                                                                                 | 11/08/1978 | Caucasian | Female | Not Hispanic |   |               |           |               | 14        | 12        | 28       | high ostomy output   | muddy brown                | Yes         | No         |
| 26 |                                                                                    |            |           |        |              |   |               |           |               |           |           |          |                      |                            |             |            |
| 27 | Note data collected from March to June 2022 while rotating through the Gastonia VA |            |           |        |              |   |               |           |               |           |           |          |                      |                            |             |            |
| 28 | Permission from Attending doctor                                                   |            |           |        |              |   |               |           |               |           |           |          |                      |                            |             |            |

# Data cleaning in action

Remember to have tidy rectangle data BEFORE you import into your chosen software, so first delete rows 1 to 4 and 27 to 28. Then, once in your chosen software, clean each variable column one at a time by looking at their distribution.

|    | A                                                                                  | B          | C         | D      | E            | Remember to have tidy rectangle data BEFORE you import into your chosen software, so first delete rows 1 to 4 and 27 to 28. Then, once in your chosen software, clean each variable column one at a time by looking at their distribution. |     |    |            |    |    |    |                      |                            |            | P   |
|----|------------------------------------------------------------------------------------|------------|-----------|--------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|------------|----|----|----|----------------------|----------------------------|------------|-----|
| 1  | Acute Kidney Injury (AKI) Study                                                    |            |           |        |              |                                                                                                                                                                                                                                            |     |    |            |    |    |    |                      |                            |            |     |
| 2  | A. Motivated Resident                                                              |            |           |        |              |                                                                                                                                                                                                                                            |     |    |            |    |    |    |                      |                            |            |     |
| 3  | Research Project                                                                   |            |           |        |              |                                                                                                                                                                                                                                            |     |    |            |    |    |    |                      |                            |            |     |
| 4  |                                                                                    |            |           |        |              |                                                                                                                                                                                                                                            |     |    |            |    |    |    |                      |                            |            |     |
| 5  | study_id                                                                           | dob        | Race      | Sex    | Hispanic     |                                                                                                                                                                                                                                            |     |    |            |    |    |    |                      | ter                        | AV fistula |     |
| 6  | 1                                                                                  | 11/11/1967 | White     | Male   | Hispanic     |                                                                                                                                                                                                                                            |     |    |            |    |    |    |                      |                            | No         |     |
| 7  | 2                                                                                  | 11/12/1990 | Caucasian | Female | Not Hispanic | 21                                                                                                                                                                                                                                         | 17  | 19 | but poor c | 12 | 19 | 26 | sepsis               | muddy brown casts          | Yes        | No  |
| 8  | 3                                                                                  | may 5 1970 | White     | Male   | Hispanic     | 32                                                                                                                                                                                                                                         | 20  | 15 |            | 11 | 15 | 24 | bleeding out         | many epithelial cell casts | No         | No  |
| 9  | 3                                                                                  | 3-Jun-81   | Other     | Female | Not Hispanic | 11                                                                                                                                                                                                                                         | 9   | 6  |            | 5  | 9  | 13 | norovirus diarrhea   | muddy brown casts          | YES        | YES |
| 10 | 4                                                                                  | 7/06/1984  | Af-Am     | Male   | Not Hispanic | 15                                                                                                                                                                                                                                         | 12  | 9  |            | 8  | 11 | 14 | sepsis               | muddy brown casts          | YES        | No  |
| 11 | 5                                                                                  | 9/04/1980  | Black     | Female | H            | 33                                                                                                                                                                                                                                         | 23  | 17 |            | 26 | 38 | 44 | liver failure        | many epithelial cell casts | No         | No  |
| 12 | 6                                                                                  | 13/01/1985 | White     | Female | Hispannic    | 26                                                                                                                                                                                                                                         | 24  | 19 |            | 33 | 39 | 43 | pneumonia sepsis     | many granular casts        | No         | No  |
| 13 | 7                                                                                  | 15/11/1968 | Asian     | F      | Hispanic     | 43                                                                                                                                                                                                                                         | 21  | 26 |            | 38 | 47 | 51 | line sepsis          | many epithelial cell casts | No`        | NO  |
| 14 | 8                                                                                  | 22/04/1982 | Mixed     | Male   | Not Hispanic | 55                                                                                                                                                                                                                                         | 24  | 28 |            | 39 | 36 | 41 | C diff diarrhea      | many epithelial cell casts | No         | No  |
| 15 | 8                                                                                  | 12/10/2003 | White     | M      | Not Hispanic | 23                                                                                                                                                                                                                                         | 11  | 6  |            | 9  | 11 | 14 | NSAID tox            | muddy brown casts          | Yes        | NO  |
| 16 | 11                                                                                 | 6/11/1982  | White     | F      | NH           | 18                                                                                                                                                                                                                                         | 17  | 24 |            | 29 | 36 | 41 | burns                | muddy brown casts          | Yes        | Yes |
| 17 | 12                                                                                 | 9/07/1979  | Caucasian | M      | Not Hispanic | 21                                                                                                                                                                                                                                         | 20  | 28 |            | 37 | 44 | 48 | pyelo                | many epithelial cell casts | No         | No  |
| 18 | 13                                                                                 | 5=11-1984  | Black     | F      | NH           | 19                                                                                                                                                                                                                                         | 20  | 23 |            | 33 | 39 | 44 | urethral obstruction | many epi cell casts        | Yes        | No  |
| 19 | 14                                                                                 | 7/04/2004  | Af-Am     | M      | Hispanic     | 26                                                                                                                                                                                                                                         | 28  | 32 |            | 41 | 44 | 49 | GI bleed             | many granular casts        | No         | No  |
| 20 | 15                                                                                 | 8/03/1990  | Asian     | M      | NH           | 36                                                                                                                                                                                                                                         | 322 | 28 |            | 36 | 39 | 43 | sepsis               | rare gran casts            | No         | No  |
| 21 | 16                                                                                 | 6/09/1983  | WHITE     | Male   | Not Hispanic | 27                                                                                                                                                                                                                                         | 24  | 29 |            | 39 | 47 | 51 | salmonella           | many granular casts        | No         | No  |
| 22 | 17                                                                                 | 11/07/1986 | WHITE     | Female | Not Hispanic | 39                                                                                                                                                                                                                                         | 32  | 37 |            | 34 | 48 | 52 | sepsis               | rare granular casts        | No         | NO  |
| 23 | 19                                                                                 | 9/06/1987  | Black     | Male   | Hispanic     | 22                                                                                                                                                                                                                                         | 20  | 24 |            | 24 | 39 | 43 | esophageal varices   | many epithelial cell casts | Yes        | No  |
| 24 | 20                                                                                 | 29/03/1979 | Caucasian | Female | Hisspanic    | 232                                                                                                                                                                                                                                        | 19  | 15 |            | 18 | 28 | 33 | dehydration          | many granular casts        | No         | No  |
| 25 | 21                                                                                 | 11/08/1978 | Caucasian | Female | Not Hispanic | 177                                                                                                                                                                                                                                        | 14  | 12 |            |    | 28 | 35 | high ostomy output   | muddy brown                | Yes        | No  |
| 26 |                                                                                    |            |           |        |              |                                                                                                                                                                                                                                            |     |    |            |    |    |    |                      |                            |            |     |
| 27 | Note data collected from March to June 2022 while rotating through the Gastonia VA |            |           |        |              |                                                                                                                                                                                                                                            |     |    |            |    |    |    |                      |                            |            |     |
| 28 | Permission from Attending doctor                                                   |            |           |        |              |                                                                                                                                                                                                                                            |     |    |            |    |    |    |                      |                            |            |     |

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# Variable types are important!



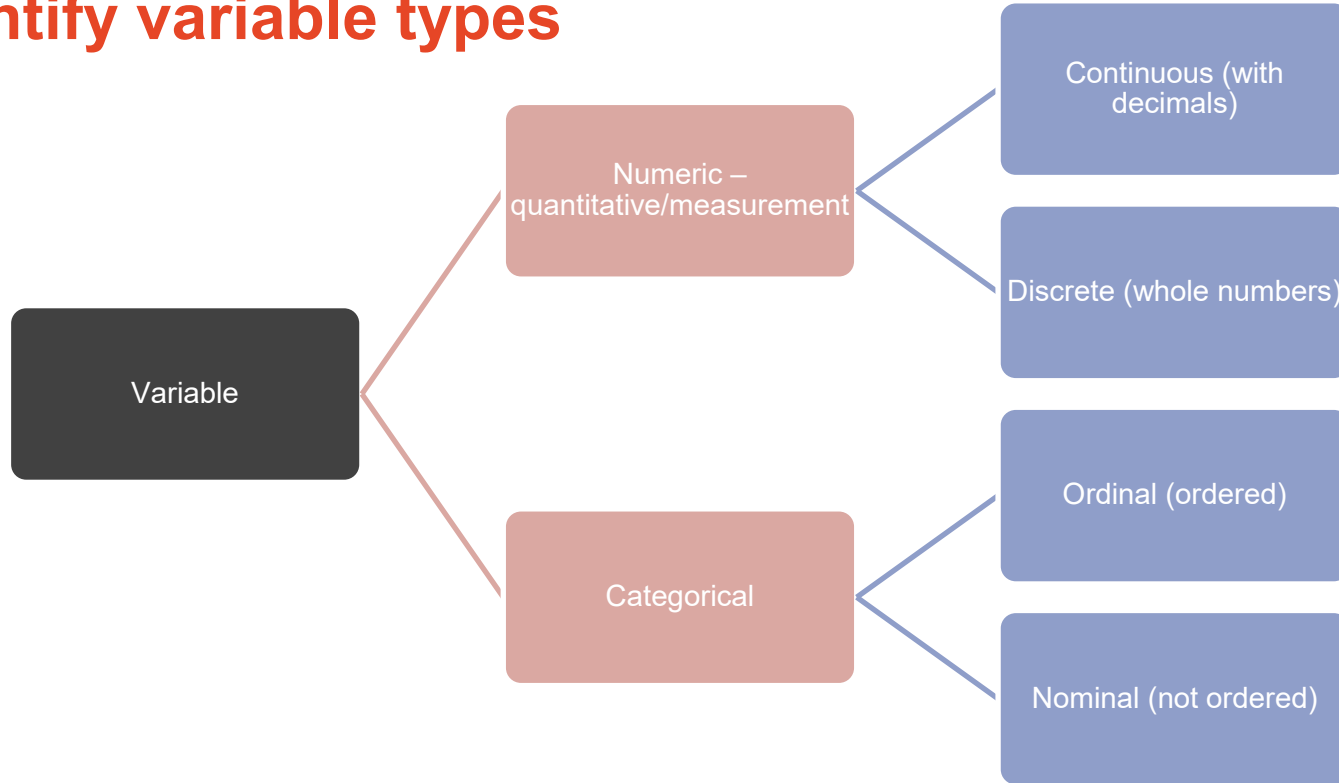
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## Why worry about variable types?

- **Variable types determine the appropriate statistical methods for analysis.**
- You need to know what data type your variable is AND how it is recorded in your data.
- You may need to convert a numeric variable to a categorical variable depending on its distribution.

---

## A. Identify variable types



# Variable types

## CONTINUOUS

measured data, can have  $\infty$  values within possible range.



I AM 3.1" TALL  
I WEIGH 34.16 grams

## DISCRETE

OBSERVATIONS can only exist at LIMITED VALUES, often COUNTS.



I HAVE 8 LEGS  
and  
4 SPOTS!

@allison-horst

# Variable types

## NOMINAL

UNORDERED DESCRIPTIONS



## ORDINAL

ORDERED DESCRIPTIONS



## BINARY

ONLY 2 MUTUALLY EXCLUSIVE OUTCOMES



@allison\_horst

---

## Variable types – Example Cath dataset

- Dataset from the Duke University Disease Databank – an observational cohort study.
- Patients were referred to the clinic for chest pain and cardiac catheterisation was performed to diagnose and open blockages in the arteries, followed by stenting to keep them open.
- 3504 participants

### Cath Dataset Description

Document: This dataset was not published as a manuscript but has been generously provided by Frank Harrell.



## Variable types – Example Cath dataset

### Codebook

| variable | position | Variable_Label                                            | units                    | Codes                | type   |
|----------|----------|-----------------------------------------------------------|--------------------------|----------------------|--------|
| sex      | 1        | Gender                                                    |                          | 0 = male, 1 = female | double |
| age      | 2        | Age                                                       | years                    |                      | double |
| cad_dur  | 3        | duration of cardiac symptoms                              | days                     |                      | double |
| choleste | 4        | Serum Cholesterol                                         | milligrams per deciliter |                      | double |
| sigdz    | 5        | Significant Coronary Artery Disease Found on Cardiac Cath |                          | 0 = no, 1 = yes      | double |
| tvdlm    | 6        | Three Vessel or Left Main Disease Found on Cardiac Cath   |                          | 0 = no, 1 = yes      | double |

---

## B. Descriptive analysis for individual variables

### Categorical variables

#### **Graphical summaries**

- Bar charts

#### **Tabular summaries**

- Frequency tables

### Numerical variables

#### **Graphical summaries**

- Histogram
- Boxplots

#### **Tabular summaries**

- Numerical summary statistics of centre (mean, median mode) and spread (quartiles, percentiles, sd)

## Example Cath dataset – How to summarise categorical variables?

## Cath Dataset

## Description

Document: This dataset was not published as a manuscript but has been generously provided by Frank Harrell.

Frequency – count the number of males and females

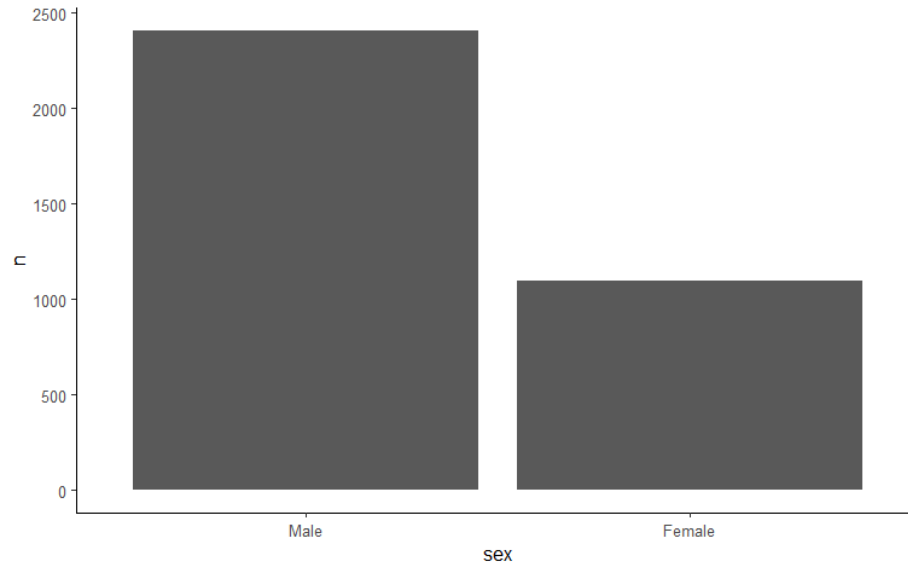
| patient_id | sex | age | age [years] | 1st Main Disease by Cardiac Cath |
|------------|-----|-----|-------------|----------------------------------|
| 1          | 0   | 73  | 132         | 268                              |
| 2          | 0   | 68  | 85          | 120                              |
| 3          | 0   | 54  | 45          | NA                               |
| 4          | 1   | 58  | 86          | 245                              |
| 5          | 1   | 56  | 7           | 269                              |
| 6          | 0   | 64  | 0           | NA                               |
| 7          | 0   | 65  | 76          | NA                               |
| 8          | 0   | 41  | 15          | 247                              |
| 9          | 0   | 68  | 30          | NA                               |
| 10         | 0   | 52  | 1           | NA                               |
| 11         | 0   | 48  | 1           | NA                               |
| 12         | 0   | 35  | 44          | 257                              |
| 13         | 1   | 69  | 10          | NA                               |
| 14         | 0   | 58  | 7           | 168                              |
| 15         | 0   | 81  | 2           | 246                              |
| 16         | 0   | 58  | 79          | 221                              |
| 17         | 0   | 59  | 36          | NA                               |
| 18         | 0   | 47  | 6           | 272                              |
| 19         | 0   | 66  | 8           | 257                              |
| 20         | 0   | 48  | 69          | 236                              |

# Example Cath dataset – How to summarise categorical variables?

- Frequency – count the number of males and females

| sex | n    | percent   |
|-----|------|-----------|
| 0   | 2405 | 0.6863584 |
| 1   | 1099 | 0.3136416 |

Where 0 = male, and 1 = female



# Example Cath dataset – How to summarise numerical variables?

- Serum cholesterol (mg per dl) of the participants who provided a response
- A frequency table would be long and messy. Not a great summary!

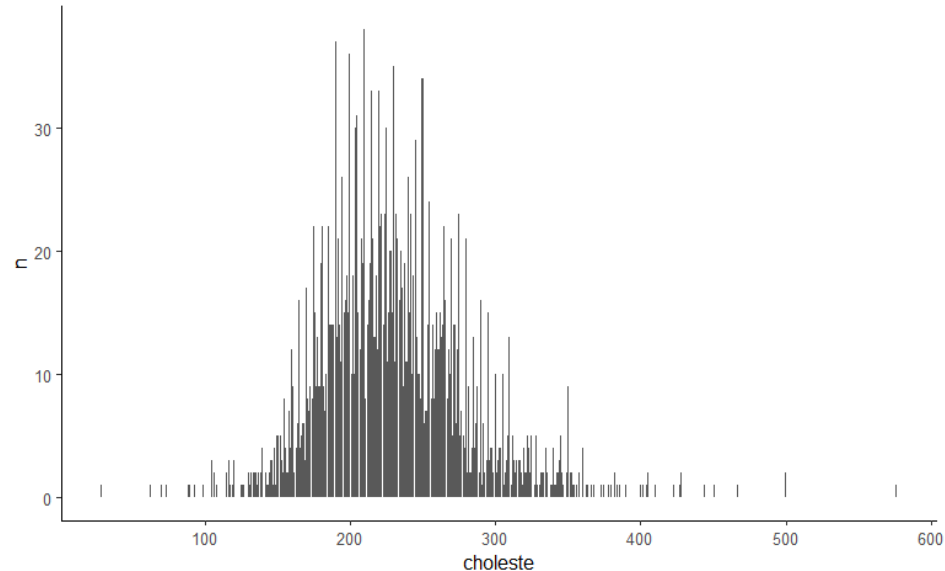
|    | choleste | n | percent      | valid_percent |
|----|----------|---|--------------|---------------|
| 1  | 29       | 1 | 0.0002853881 | 0.0004428698  |
| 2  | 63       | 1 | 0.0002853881 | 0.0004428698  |
| 3  | 70       | 1 | 0.0002853881 | 0.0004428698  |
| 4  | 74       | 1 | 0.0002853881 | 0.0004428698  |
| 5  | 89       | 1 | 0.0002853881 | 0.0004428698  |
| 6  | 90       | 1 | 0.0002853881 | 0.0004428698  |
| 7  | 93       | 1 | 0.0002853881 | 0.0004428698  |
| 8  | 99       | 1 | 0.0002853881 | 0.0004428698  |
| 9  | 105      | 3 | 0.0008561644 | 0.0013286094  |
| 10 | 107      | 2 | 0.0005707763 | 0.0008857396  |
| 11 | 108      | 1 | 0.0002853881 | 0.0004428698  |
| 12 | 115      | 2 | 0.0005707763 | 0.0008857396  |
| 13 | 117      | 3 | 0.0008561644 | 0.0013286094  |
| 14 | 118      | 1 | 0.0002853881 | 0.0004428698  |
| 15 | 119      | 1 | 0.0002853881 | 0.0004428698  |
| 16 | 120      | 3 | 0.0008561644 | 0.0013286094  |
| 17 | 125      | 1 | 0.0002853881 | 0.0004428698  |
| 18 | 126      | 1 | 0.0002853881 | 0.0004428698  |
| 19 | 127      | 1 | 0.0002853881 | 0.0004428698  |
| 20 | 130      | 2 | 0.0005707763 | 0.0008857396  |
| 21 | 131      | 1 | 0.0002853881 | 0.0004428698  |
| 22 | 132      | 2 | 0.0005707763 | 0.0008857396  |
| 23 | 134      | 2 | 0.0005707763 | 0.0008857396  |
| 24 | 135      | 2 | 0.0005707763 | 0.0008857396  |
| 25 | 136      | 1 | 0.0002853881 | 0.0004428698  |
| 26 | 137      | 2 | 0.0005707763 | 0.0008857396  |
| 27 | 139      | 2 | 0.0005707763 | 0.0008857396  |
| 28 | 140      | 4 | 0.0011415525 | 0.0017714792  |
| 29 | 142      | 2 | 0.0005707763 | 0.0008857396  |
| 30 | 143      | 1 | 0.0002853881 | 0.0004428698  |
| 31 | 144      | 1 | 0.0002853881 | 0.0004428698  |

|     | choleste | n    | percent      | valid_percent |
|-----|----------|------|--------------|---------------|
| 234 | 353      | 2    | 0.0005707763 | 0.0008857396  |
| 235 | 354      | 1    | 0.0002853881 | 0.0004428698  |
| 236 | 356      | 1    | 0.0002853881 | 0.0004428698  |
| 237 | 358      | 2    | 0.0005707763 | 0.0008857396  |
| 238 | 360      | 4    | 0.0011415525 | 0.0017714792  |
| 239 | 363      | 1    | 0.0002853881 | 0.0004428698  |
| 240 | 364      | 1    | 0.0002853881 | 0.0004428698  |
| 241 | 366      | 1    | 0.0002853881 | 0.0004428698  |
| 242 | 368      | 1    | 0.0002853881 | 0.0004428698  |
| 243 | 373      | 1    | 0.0002853881 | 0.0004428698  |
| 244 | 375      | 1    | 0.0002853881 | 0.0004428698  |
| 245 | 378      | 1    | 0.0002853881 | 0.0004428698  |
| 246 | 380      | 1    | 0.0002853881 | 0.0004428698  |
| 247 | 382      | 2    | 0.0005707763 | 0.0008857396  |
| 248 | 384      | 1    | 0.0002853881 | 0.0004428698  |
| 249 | 386      | 1    | 0.0002853881 | 0.0004428698  |
| 250 | 390      | 1    | 0.0002853881 | 0.0004428698  |
| 251 | 400      | 1    | 0.0002853881 | 0.0004428698  |
| 252 | 402      | 1    | 0.0002853881 | 0.0004428698  |
| 253 | 404      | 1    | 0.0002853881 | 0.0004428698  |
| 254 | 405      | 2    | 0.0005707763 | 0.0008857396  |
| 255 | 410      | 1    | 0.0002853881 | 0.0004428698  |
| 256 | 423      | 1    | 0.0002853881 | 0.0004428698  |
| 257 | 427      | 1    | 0.0002853881 | 0.0004428698  |
| 258 | 428      | 2    | 0.0005707763 | 0.0008857396  |
| 259 | 444      | 1    | 0.0002853881 | 0.0004428698  |
| 260 | 451      | 1    | 0.0002853881 | 0.0004428698  |
| 261 | 467      | 1    | 0.0002853881 | 0.0004428698  |
| 262 | 500      | 2    | 0.0005707763 | 0.0008857396  |
| 263 | 576      | 1    | 0.0002853881 | 0.0004428698  |
| 264 | N/A      | 1246 | 0.3555936073 | N/A           |

---

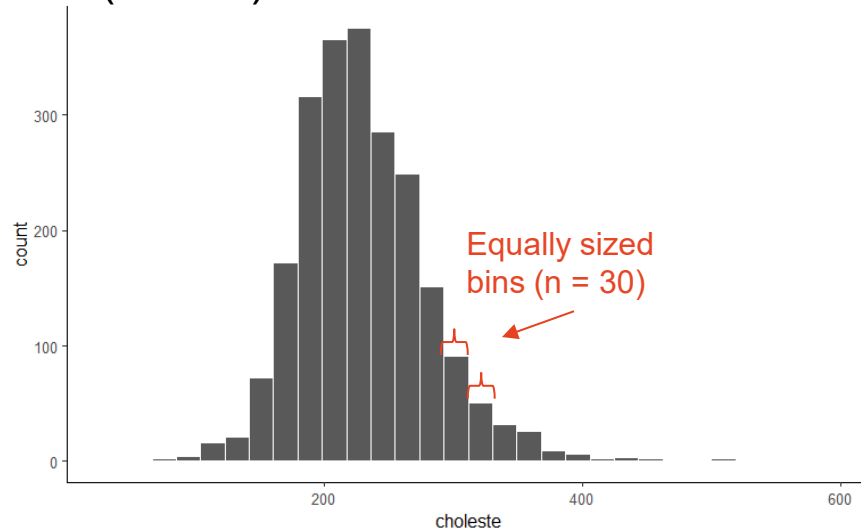
## Example Cath dataset – How to summarise numerical variables?

- **Serum cholesterol (mg per dl) of the participants who provided a response**
- A bar chart isn't the best either! 😞



## Example Cath dataset – How to summarise numerical variables?

- **Serum cholesterol (mg per dl) of the participants who provided a response**
- A histogram of cholesterol, with frequency shown with equally sized bins ( $n = 30$ ).





## Example Cath dataset – How to summarise numerical variables?

- **Age (years) of the participants who provided a response**
- A frequency table would be long and messy. Not a great summary!

|    | age | n  | percent      |
|----|-----|----|--------------|
| 1  | 17  | 1  | 0.0002853881 |
| 2  | 18  | 1  | 0.0002853881 |
| 3  | 19  | 1  | 0.0002853881 |
| 4  | 20  | 2  | 0.0005707763 |
| 5  | 22  | 2  | 0.0005707763 |
| 6  | 23  | 1  | 0.0002853881 |
| 7  | 24  | 2  | 0.0005707763 |
| 8  | 25  | 4  | 0.0011415525 |
| 9  | 26  | 2  | 0.0005707763 |
| 10 | 27  | 5  | 0.0014269406 |
| 11 | 28  | 11 | 0.0031392694 |
| 12 | 29  | 9  | 0.0025684932 |
| 13 | 30  | 10 | 0.0028538813 |
| 14 | 31  | 18 | 0.0051369863 |
| 15 | 32  | 17 | 0.0048515982 |

|    | age | n   | percent      |
|----|-----|-----|--------------|
| 16 | 33  | 15  | 0.0042808219 |
| 17 | 34  | 30  | 0.0085616438 |
| 18 | 35  | 30  | 0.0085616438 |
| 19 | 36  | 34  | 0.0097031963 |
| 20 | 37  | 46  | 0.0131278539 |
| 21 | 38  | 56  | 0.0159817352 |
| 22 | 39  | 61  | 0.0174086758 |
| 23 | 40  | 63  | 0.0179794521 |
| 24 | 41  | 84  | 0.0239726027 |
| 25 | 42  | 76  | 0.0216894977 |
| 26 | 43  | 90  | 0.0256849315 |
| 27 | 44  | 102 | 0.0291095890 |
| 28 | 45  | 95  | 0.0271118721 |
| 29 | 46  | 130 | 0.0371004566 |
| 30 | 47  | 111 | 0.0316780822 |

|    | age | n   | percent      |
|----|-----|-----|--------------|
| 31 | 48  | 134 | 0.0382420091 |
| 32 | 49  | 144 | 0.0410958904 |
| 33 | 50  | 136 | 0.0388127854 |
| 34 | 51  | 124 | 0.0353881279 |
| 35 | 52  | 116 | 0.0331050228 |
| 36 | 53  | 128 | 0.0365296804 |
| 37 | 54  | 126 | 0.0359589041 |
| 38 | 55  | 126 | 0.0359589041 |
| 39 | 56  | 156 | 0.0445205479 |
| 40 | 57  | 144 | 0.0410958904 |
| 41 | 58  | 126 | 0.0359589041 |
| 42 | 59  | 96  | 0.0273972603 |
| 43 | 60  | 99  | 0.0282534247 |
| 44 | 61  | 97  | 0.0276826484 |
| 45 | 62  | 78  | 0.0222602740 |

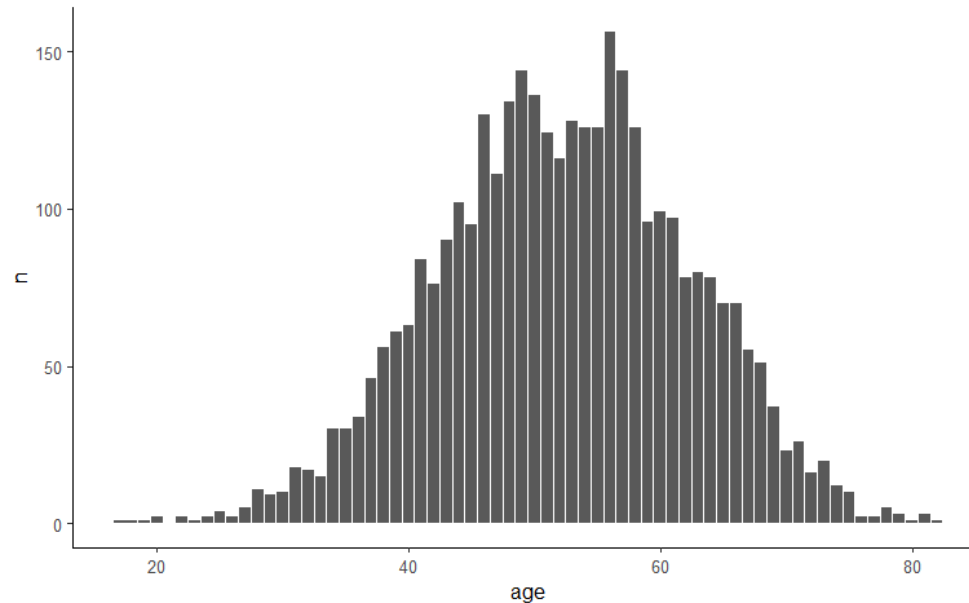
|    | age | n  | percent      |
|----|-----|----|--------------|
| 46 | 63  | 80 | 0.0228310502 |
| 47 | 64  | 78 | 0.0222602740 |
| 48 | 65  | 70 | 0.0199771689 |
| 49 | 66  | 70 | 0.0199771689 |
| 50 | 67  | 55 | 0.0156963470 |
| 51 | 68  | 51 | 0.0145547945 |
| 52 | 69  | 37 | 0.0105593607 |
| 53 | 70  | 23 | 0.0065639269 |
| 54 | 71  | 26 | 0.0074200913 |
| 55 | 72  | 16 | 0.0045662100 |
| 56 | 73  | 20 | 0.0057077626 |
| 57 | 74  | 12 | 0.0034246575 |
| 58 | 75  | 10 | 0.0028538813 |
| 59 | 76  | 2  | 0.0005707763 |
| 60 | 77  | 2  | 0.0005707763 |

|    | age | n | percent      |
|----|-----|---|--------------|
| 61 | 78  | 5 | 0.0014269406 |
| 62 | 79  | 3 | 0.0008561644 |
| 63 | 80  | 1 | 0.0002853881 |
| 64 | 81  | 3 | 0.0008561644 |
| 65 | 82  | 1 | 0.0002853881 |



## Example Cath dataset – How to summarise numerical variables?

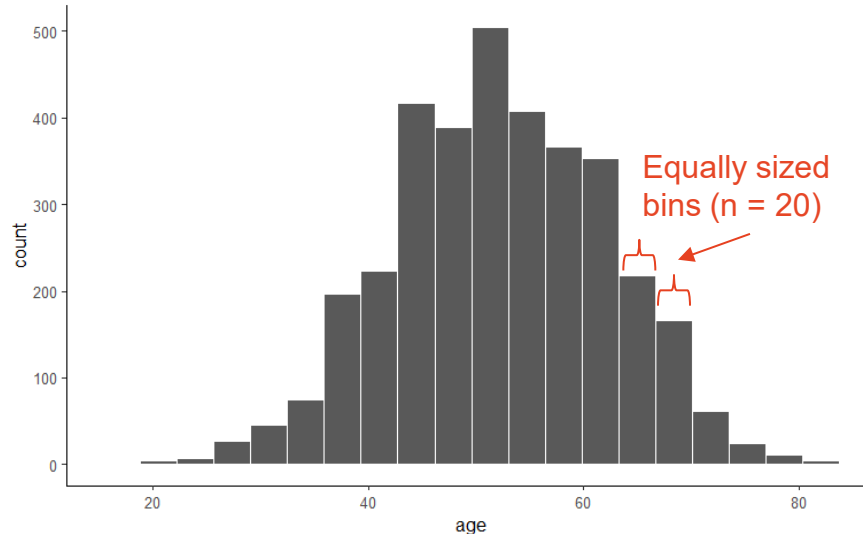
- **Age (years) of the participants who provided a response**
- A bar chart isn't the best either!



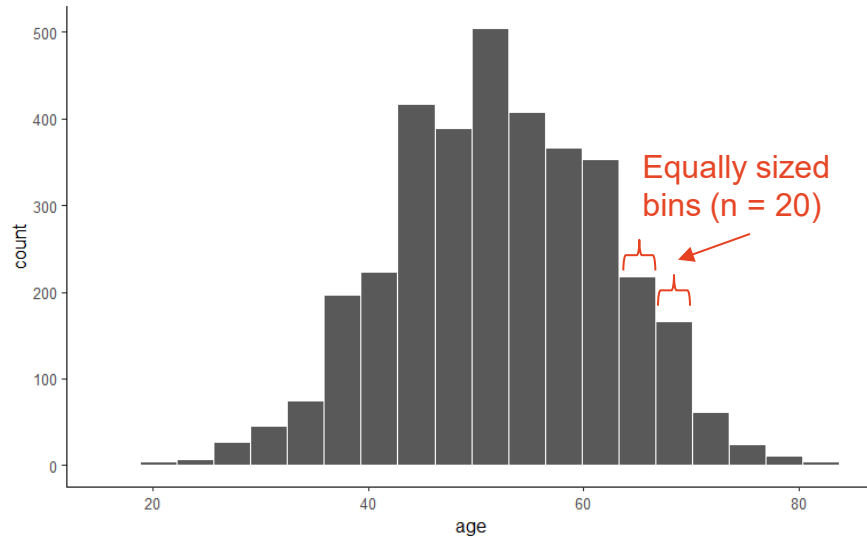


## Example Cath dataset – How to summarise numerical variables?

- **Age (years) of the participants who provided a response**
- A histogram of age, with frequency shown with equally sized bins ( $n = 20$ ).

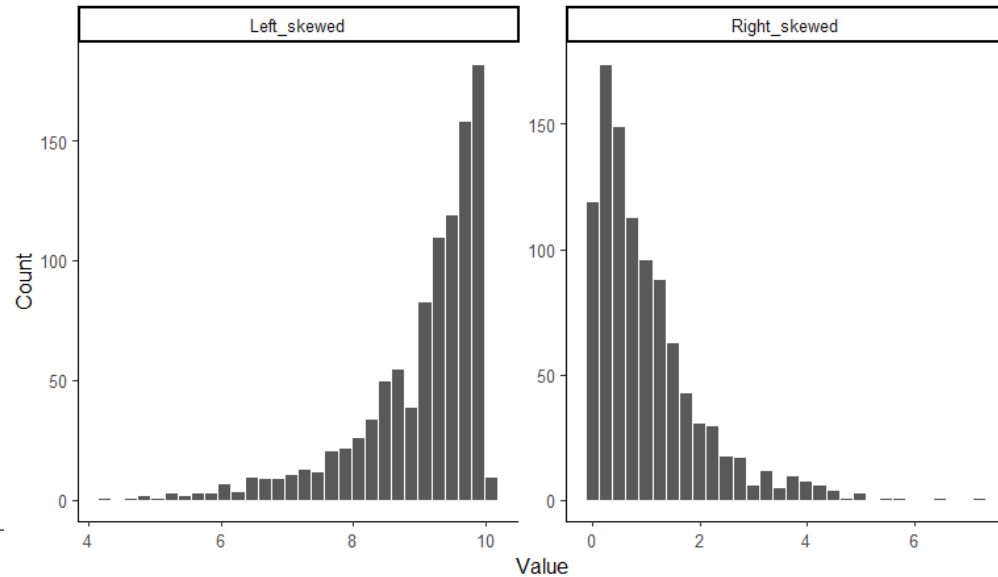


# Shapes of the distribution



Symmetric distribution

Examples of Skewed Distributions

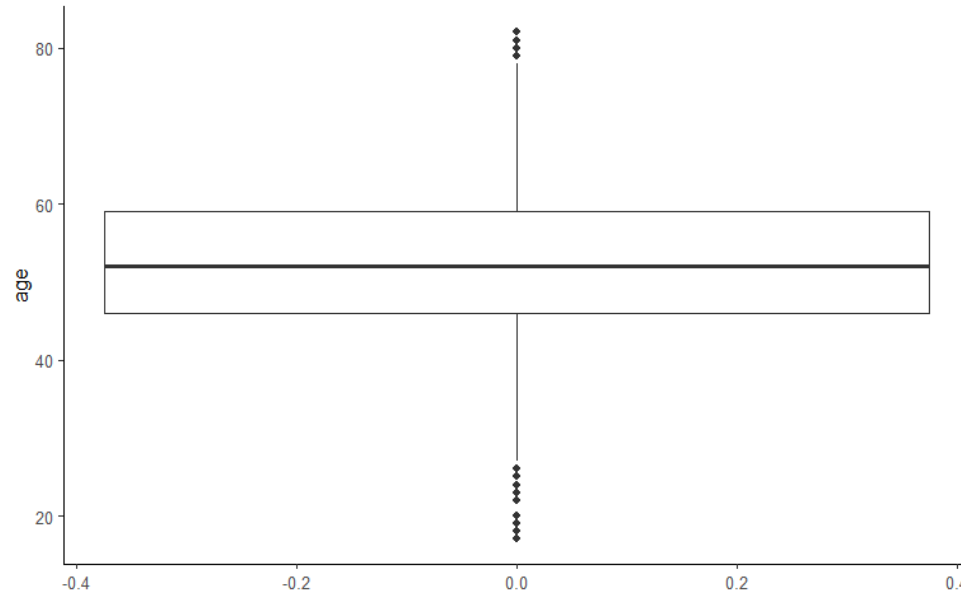


Asymmetric distribution

---

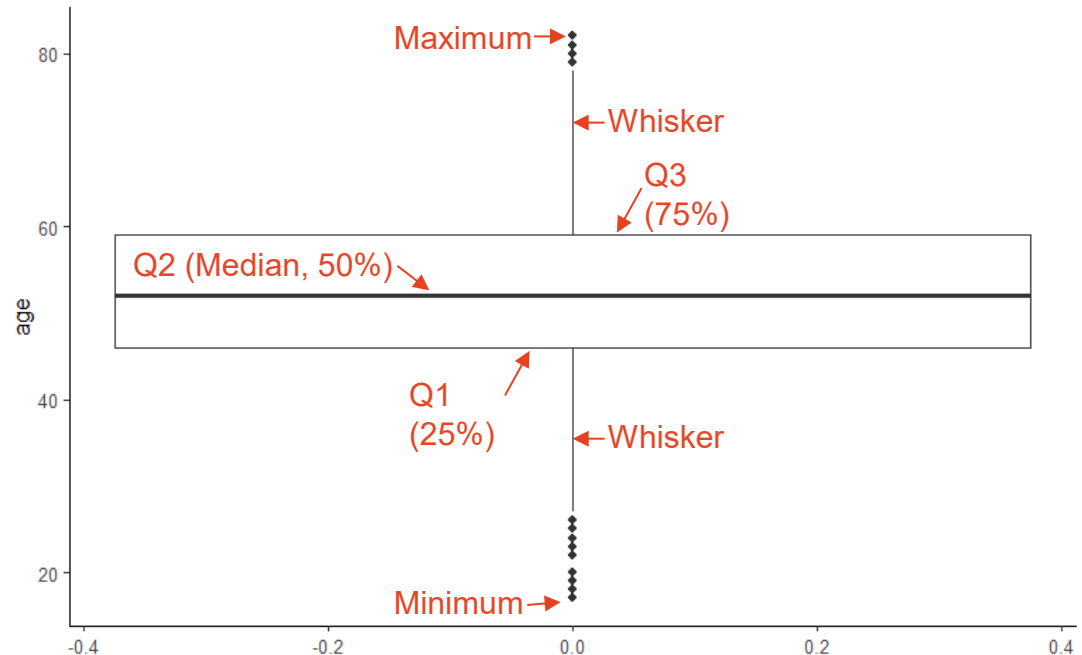
## Example Cath dataset – How to summarise numerical variables?

- Another way to summarise a numerical variable is through a boxplot.

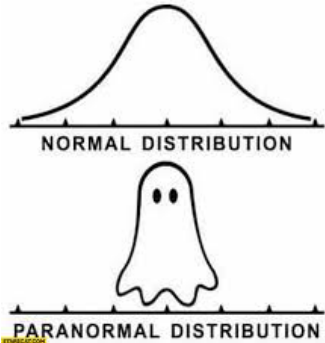


## Example Cath dataset – Summarising the boxplot

- **A numerical distribution can be summarised by giving descriptions of:**
  - Its shape
    - Symmetric, right or left skewed
  - Its centre
    - Measures of central value or location
  - Its spread
    - Measures of spread or dispersion



# Parametric versus Nonparametric stats



## - **Parametric**

- Based on assumptions about data distribution and shape.
  - Normality
    - Though quite robust in large samples due to the central limit theorem.
  - Homogeneity of variance.
- Mean and standard deviation reported.
- Suitable for larger samples, with a normal distribution. More powerful if assumptions met.
- Examples:
  - 2-sample t-test
  - General linear model (GLM, One-way Anova, Least squares regression)

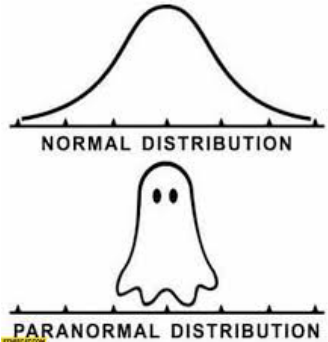
## - **Nonparametric**

- Not based on assumptions about data distribution and shape.
- Median and percentiles/quartiles/max and min reported.
- Suitable for smaller samples, skewed data or ordinal variables.
- Examples:
  - Mann-Whitney U
  - Kruskal-Wallis
  - Spearman correlation

*With the rise of GLMs and other more flexible parametric methods, such as distributional regression, quantile regression and generalised additive models, nonparametric methods are becoming less useful (and are now less of a requirement for publication).*



# Improving data distribution with transformations



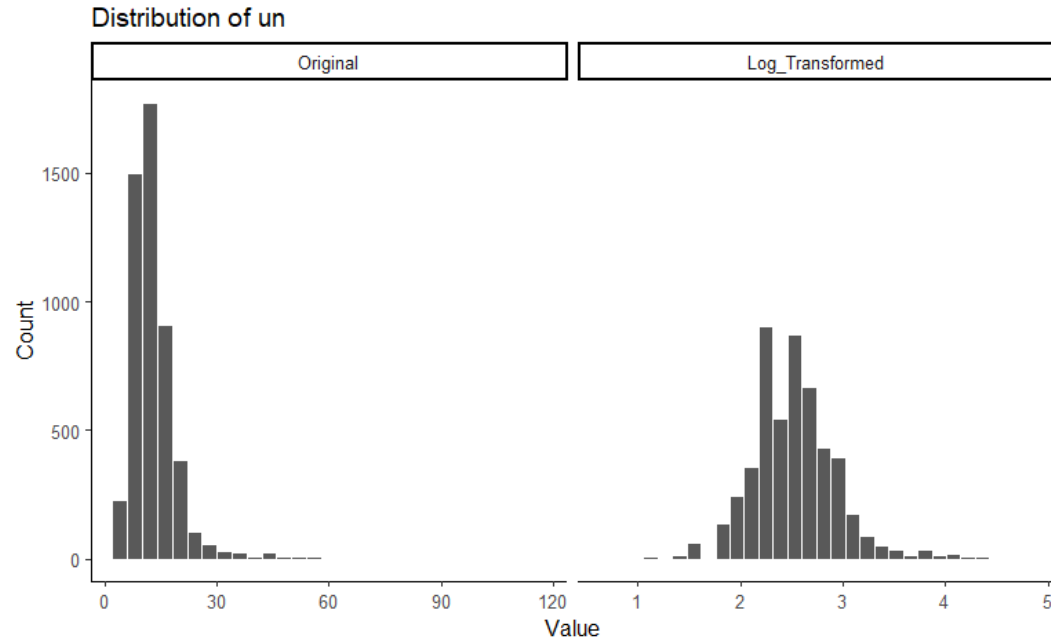
- **Why check the distribution?**
  - Helps us understand skewness and any extreme values in the raw data.
- **Common transformations:** log, square root, inverse, Box-Cox.
  - We cannot take a log of 0, so if the raw data contains values of 0, add a constant to the variable before taking the log. E.g.  $\log(x + 1)$ .
- Checking the raw data is an important part of EDA. However, when fitting linear models, we assess distributional assumptions using the residuals, not the raw data. See the [Linear Models 1](#) workshop for more information.

[Best practice in statistics: The use of log transformation](#)



# Improving data distribution with transformations

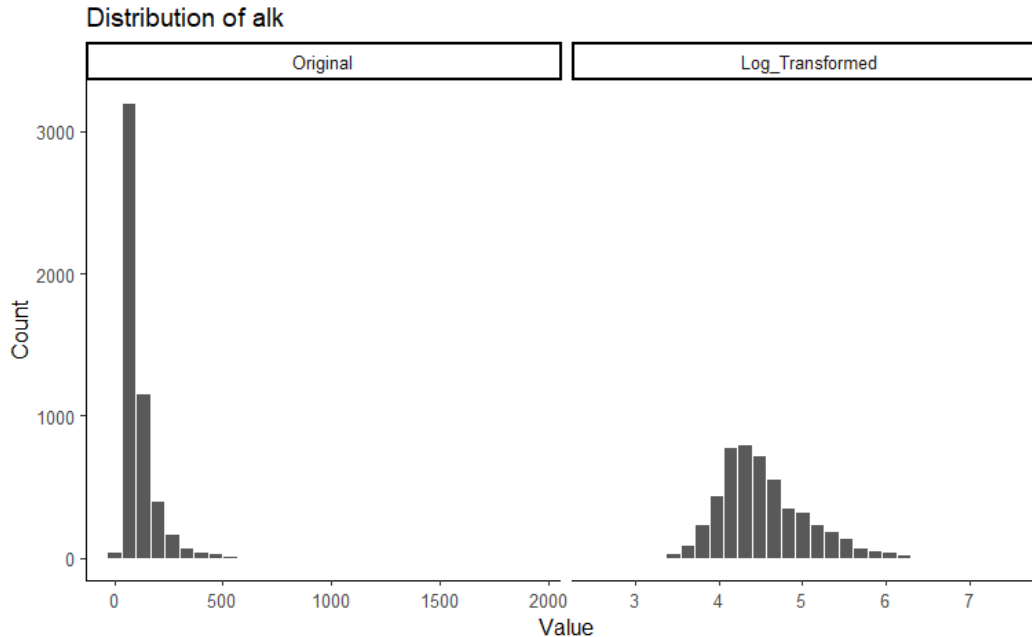
- **UN:** Blood Urea Nitrogen. Numeric. Range: 2-118. 53 missing values.





# Improving data distribution with transformations

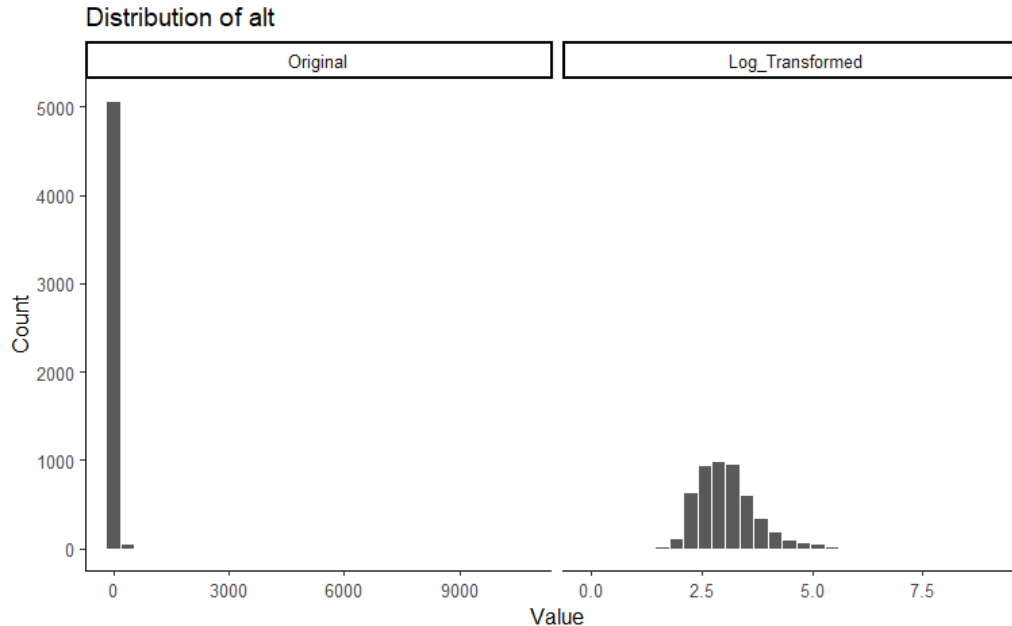
- **ALK:** Alkaline phosphatase. Numeric, range 13-1938, 0 missing values.





# Improving data distribution with transformations

- **ALT:** Alanine Transaminase. Numeric, range 1-10666, 18 missing values.



---

# What is a systematic approach to conduct descriptive analyses for individual variables?

## Categorical variables

### Graphical summaries

- Bar charts

### Tabular summaries

- Frequency tables

**Don't forget to check for missing data/NAs!**

For formatting tips, check your target journal's instructions to authors or recent publications in that journal!

## Numerical variables

### Graphical summaries

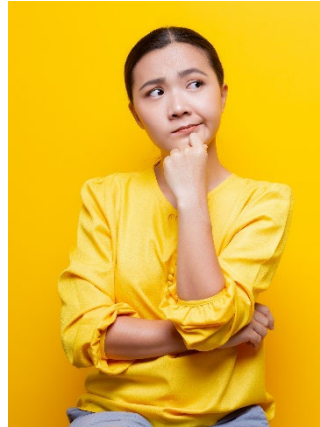
- Histogram
- Boxplots

### Tabular summaries

- Symmetric?
  - Mean, standard deviation, minimum and maximum.
- Asymmetric?
  - Median, quartiles, minimum and maximum.

---

**Variable types are important for describing the distribution of each variable and checking/cleaning each individual variable, but....**



**What about my research question and analysis?**

# What is the research question?

Depending on who taught you statistics, your discipline, stats software or textbook, you may have come across some of these terms and more!

- For any analysis we need to be clear what the functional classification of the variables in the dataset is, e.g. we want to investigate the effect of smoking on lung disease:



**Smoking (yes/no)**  
Predictor, explanatory  
variable, risk factor,  
independent variable\*



**Lung disease (yes/no)**  
Response, outcome,  
dependent variable

\*This term is frequently used, but we don't promote it as predictors may be correlated and hence are not independent.

---

## Other functional classifications for variable types

- Covariate: a measured predictor (numerical predictor variable).
- Factor: (a categorical predictor variable).

### **Experimental design variables:**

- Design variables: Based on the physical design of the experiment. They are often included in the analysis even if not 'significant' e.g., correctly partition the variance e.g., Block (batch of reagent, source of lab mice), subject ID, etc.
- Treatment: Variables of interest, e.g., diet, drug treatment, intervention etc. NB: The 'levels' of a 'treatment variable' might include 'control (placebo)', 'treatment 1 (drug 1)', 'treatment 2 (drug 2)'.

**More information on design variables in our Experimental Design workshop!**

---

# What is the outcome variable?

- **Review study aim and objectives**
- E.g., vaccine RCT - daily morbidity outcome data could be analysed as:
  - mean daily rate (average – numerical).
  - cumulative morbidity (sum – numerical).
  - peak morbidity (maximum – numerical).
  - outbreak presence/absence (binary group – categorical).
  - time to infection/disease outbreak (time to binary event data – survival analysis).

---

## More data processing

- **Assess all variables for missing observations – if many missing consider analysing with and without that predictor.**
- **Check the distribution of all variables individually (previous step).**
- Numerical predictors: handle as numerical or categorical?
- Categorical: may have to combine categories if there are low frequency counts (if it makes sense to do so).
- **Multi-level (clustered) data**
- Each observation/row uniquely identified? E.g., herd, animal, ID.
- Evaluate hierarchical structure of your data: Average/range of observations at one level in each higher level?
- E.g., mean, min, max of students/class; mean, min, max of classes/school.

---

# A quick primer to Step 5: EDA

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# Your variable types will dictate the type of statistical analysis you perform

- The type of outcome and explanatory variables you have will dictate the type of EDA and inferential analysis you can do, so it is crucial to think about this and your research question before you collect your data!
  - Is your outcome numerical or categorical?
  - Are your explanatory variables numerical or categorical?
  - Do you have one or multiple outcomes and/or explanatory variables?
- Choosing the wrong analysis can violate statistical assumptions and often won't run properly in your statistical software. Even worse, the incorrect analysis could lead to erroneous results and therefore conclusions or policy!

## Step 5: Exploratory data analysis (EDA)



It will depend on the analysis and variables involved.



Basic EDA is where you plot the relationship of each predictor with the outcome, and you may need to reconsider data processing. E.g. Do I need to merge more categories together?

### Two categorical variables

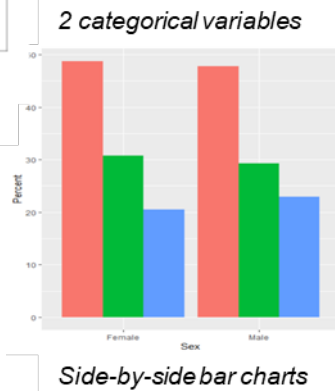
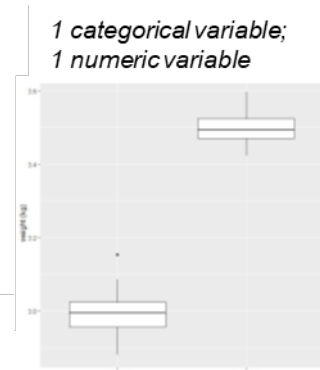
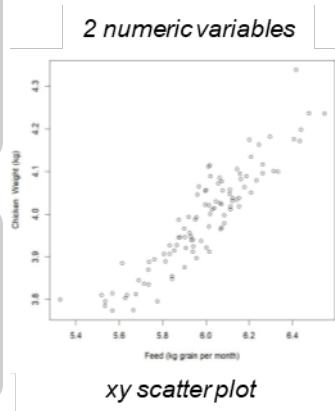
- Contingency table
- Side-by-side bar chart

### A categorical and a numerical variable

- Tabulate summary statistics by groups
- Box-and-whisker plot by groups

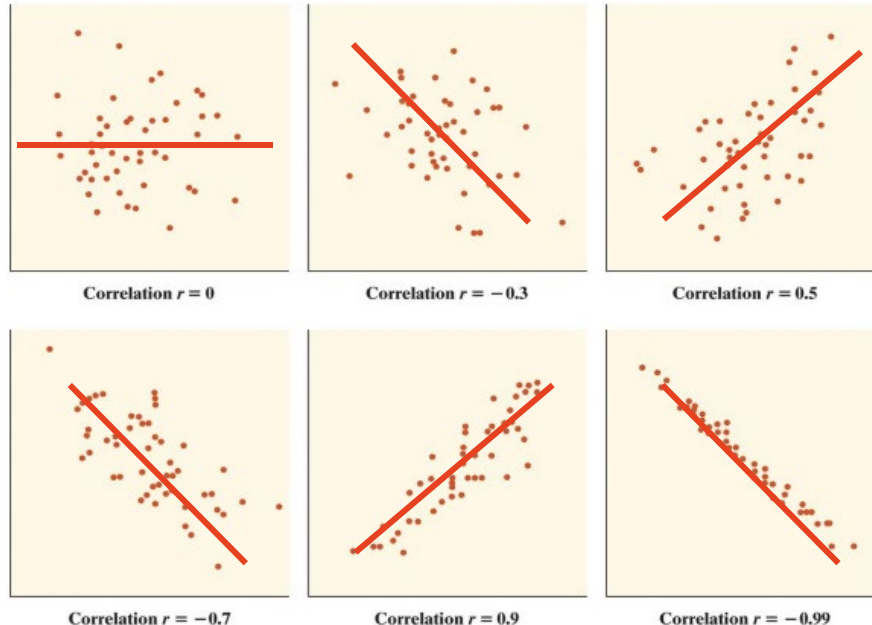
### Two numerical variables

- Scatter plot and correlation coefficient  $r$



# XY scatterplot and Pearson's correlation coefficient ( $r$ )

A quick review of correlation coefficient  $r$  to describe the relationship of two numerical variables in a scatter plot;  $r = 0$  means no relationship – data points in a random scatter.



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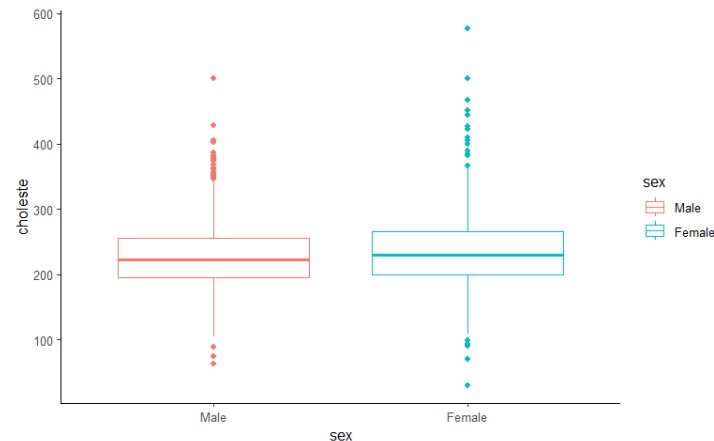
## Examples for reporting descriptive analyses:

- Plotting gives a quick visual summary during EDA and highlights any issues.
- Tables are more publication friendly as they save space!
  - This step of the statistical analysis process is often seen in Table 1 of your manuscript.
  - See this paper for more information: [Who is in this study, anyway? Guidelines for a useful Table 1.](#)

# Examples for reporting descriptive analyses: Cath dataset

**Table 1.** Summary statistics for the variables in a study of patients undergoing cardiac catheterisation, split according to sex.

|                                                        | Male<br>(N=2405)  | Female<br>(N=1099) | Overall<br>(N=3504) |
|--------------------------------------------------------|-------------------|--------------------|---------------------|
| <b>Age [Year]</b>                                      |                   |                    |                     |
| Mean (SD)                                              | 51.6 (9.83)       | 53.8 (9.99)        | 52.3 (9.93)         |
| Median [Min, Max]                                      | 52.0 [17.0, 82.0] | 54.0 [20.0, 81.0]  | 52.0 [17.0, 82.0]   |
| <b>Duration of Symptoms of Coronary Artery Disease</b> |                   |                    |                     |
| Mean (SD)                                              | 43.0 (58.1)       | 43.0 (58.3)        | 43.0 (58.2)         |
| Median [Min, Max]                                      | 17.0 [0, 416]     | 20.0 [0, 404]      | 18.0 [0, 416]       |
| <b>Cholesterol [mg %]</b>                              |                   |                    |                     |
| Mean (SD)                                              | 227 (47.3)        | 237 (56.9)         | 230 (50.6)          |
| Median [Min, Max]                                      | 223 [63.0, 500]   | 230 [29.0, 576]    | 225 [29.0, 576]     |
| Missing                                                | 836 (34.8%)       | 410 (37.3%)        | 1246 (35.6%)        |
| <b>Significant Coronary Artery Disease</b>             |                   |                    |                     |
| No                                                     | 533 (22.2%)       | 637 (58.0%)        | 1170 (33.4%)        |
| Yes                                                    | 1872 (77.8%)      | 462 (42.0%)        | 2334 (66.6%)        |
| <b>Three Vessel or Left Main Disease</b>               |                   |                    |                     |
| No                                                     | 1463 (60.8%)      | 909 (82.7%)        | 2372 (67.7%)        |
| Yes                                                    | 941 (39.1%)       | 188 (17.1%)        | 1129 (32.2%)        |
| Missing                                                | 1 (0.0%)          | 2 (0.2%)           | 3 (0.1%)            |



[Cath Dataset Description Document:](#) This dataset was not published as a manuscript but has been generously provided by Frank Harrell.

# Other examples for reporting descriptive analyses:

## Midurethral Sling vs OnabotulinumtoxinA in Females With Urinary Incontinence

Table 1. Demographics and Baseline Characteristics

| Characteristic                                                                | No. (%)                                  | No. (%)                                 |
|-------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------|
|                                                                               | OnabotulinumtoxinA (n = 71) <sup>a</sup> | Midurethral sling (n = 66) <sup>a</sup> |
| Age, mean (SD) [range], y                                                     | 59.1 (11.4) [27-78]                      | 59.0 (11.7) [33-87]                     |
| Race <sup>b</sup>                                                             |                                          |                                         |
| Asian                                                                         | 2 (2.8)                                  | 0                                       |
| Black/African American                                                        | 10 (14.1)                                | 10 (15.2)                               |
| Native Hawaiian or Other Pacific Islander                                     | 1 (1.4)                                  | 1 (1.5)                                 |
| White                                                                         | 55 (77.5)                                | 54 (81.8)                               |
| Unknown/not reported                                                          | 3 (4.2)                                  | 1 (1.5)                                 |
| Hispanic/Latina ethnicity, No./total (%) <sup>c</sup>                         | 15/71 (21.1)                             | 6/65 (9.1)                              |
| Education, highest level obtained was greater than high school, No./total (%) | 46/68 (64.8)                             | 32/63 (48.5)                            |
| Currently smoking                                                             | 7 (9.9)                                  | 9 (13.6)                                |
| No. of vaginal deliveries, median (IQR)                                       | 2 (1-3)                                  | 2 (1-3)                                 |
| Total No. of deliveries, median (IQR)                                         | 2 (1-3)                                  | 2 (2-3)                                 |
| Menopausal status                                                             |                                          |                                         |
| Pre                                                                           | 7 (9.9)                                  | 14 (21.2)                               |
| Post                                                                          | 56 (78.9)                                | 49 (74.2)                               |
| Not sure                                                                      | 8 (11.3)                                 | 3 (4.5)                                 |
| Currently using estrogen by prescription                                      | 21 (29.6)                                | 18 (27.3)                               |
| BMI, mean (SD) <sup>d</sup>                                                   | 34.3 (8.3)                               | 35.0 (7.6)                              |
| Type of urinary incontinence <sup>e</sup>                                     |                                          |                                         |
| Stress predominant                                                            | 3 (4.2)                                  | 4 (6.1)                                 |
| Urge predominant                                                              | 7 (9.9)                                  | 10 (15.2)                               |
| Balanced                                                                      | 61 (85.9)                                | 52 (78.8)                               |
| Baseline incontinence episode daily frequency, mean (SD)                      | 7.1 (4.1)                                | 7.4 (4.0)                               |
| Time from baseline visit to treatment, mean (SD), d <sup>f</sup>              | 59.0 (38.5)                              | 56.0 (43.4)                             |
| Median (IQR)                                                                  | 49 (36-75)                               | 46 (27-76)                              |
| Treatment >90 d after baseline                                                | 7 (9.9)                                  | 10 (15.2)                               |
| Baseline UII scores, mean (SD) <sup>g</sup>                                   |                                          |                                         |
| Total                                                                         | 187.6 (38.2)                             | 180.9 (36.5)                            |
| Irritative                                                                    | 75.7 (15.6)                              | 77.3 (15.4)                             |
| Stress                                                                        | 86.6 (18.4)                              | 80.3 (21.9)                             |

Abbreviations: BMI, body mass index; UII, Urogenital Distress Inventory.

<sup>a</sup> The primary analysis population is defined as all participants who received any treatment and have postbaseline efficacy data, regardless of randomized treatment.

<sup>b</sup> Race categories were self-reported using check all that apply and specific closed options, including an unknown/not reported selection.

<sup>c</sup> Ethnicity categories were self-reported using select only one, specific closed option, including an unknown/not reported selection.

<sup>d</sup> BMI calculated as weight in kilograms divided by height in square meters.

<sup>e</sup> The type of urinary incontinence is defined by responses at baseline on the UII to the urgency urinary incontinence (UII) item "Do you experience urine leakage related to a feeling of urgency? If yes, how much does it bother you?"

and stress urinary incontinence (SUI) item "Do you experience urine leakage related to physical activity, coughing or sneezing? If yes, how much does it bother you?" Greater bother reported on the UII item is classified as stress predominant, greater bother reported on the SUI item is classified as stress predominant, and equal bother reported is classified as balanced.

<sup>f</sup> Baseline refers to the time of the first UII assessment completion, which is used to determine eligibility. Participants were expected to receive treatment within 91 days of completing their baseline UII.

<sup>g</sup> The UII total score ranges from 0 to 300, and the UII irritative and stress scores range from 0 to 100, with higher scores indicating greater symptom severity.

## Immunogenicity and Safety of Influenza and COVID-19 Multicomponent Vaccine in Adults ≥50 Years

Table 1. Participant Demographics and Baseline Characteristics (Safety Population)<sup>a</sup>

|                                                     | Age ≥65 y                       | Age 50-64 y                     |                                 |
|-----------------------------------------------------|---------------------------------|---------------------------------|---------------------------------|
|                                                     | HD-ILIV4 + mRNA-1083 (n = 2011) | SD-ILIV4 + mRNA-1273 (n = 2006) | SD-ILIV4 + mRNA-1273 (n = 1993) |
| Age, y                                              |                                 |                                 |                                 |
| Mean (SD)                                           | 70.9 (5.0)                      | 70.7 (4.7)                      | 57.5 (4.3)                      |
| Median (IQR)                                        | 70 (67-74)                      | 70 (67-74)                      | 58 (54-61)                      |
| Age group, No. (%)                                  |                                 |                                 |                                 |
| 50-64 y                                             | 2 (<0.1) <sup>b</sup>           | 0                               | 1991 (99.9)                     |
| 65-74 y                                             | 1593 (79.2)                     | 1591 (79.3)                     | 1 (<0.1) <sup>b</sup>           |
| ≥75 y                                               | 416 (20.7)                      | 415 (20.7)                      | 1 (<0.1) <sup>b</sup>           |
| Sex, No. (%)                                        |                                 |                                 |                                 |
| Male                                                | 933 (46.4)                      | 908 (45.3)                      | 837 (42.0)                      |
| Female                                              | 1078 (53.6)                     | 1098 (54.7)                     | 1156 (58.0)                     |
| Race, No. (%) <sup>c</sup>                          | n = 2000                        | n = 1994                        | n = 1978                        |
| American Indian or Alaska Native                    | 9 (0.5)                         | 12 (0.6)                        | 13 (0.7)                        |
| Asian                                               | 25 (1.3)                        | 36 (1.8)                        | 50 (2.5)                        |
| Black or African American                           | 370 (18.5)                      | 370 (18.6)                      | 517 (26.1)                      |
| Native Hawaiian or Other Pacific Islander           | 1 (0.1)                         | 4 (0.2)                         | 3 (0.2)                         |
| White                                               | 1577 (78.9)                     | 1565 (78.5)                     | 1373 (69.4)                     |
| Multiple                                            | 14 (0.7)                        | 4 (0.2)                         | 19 (1.0)                        |
| Other                                               | 4 (0.2)                         | 3 (0.2)                         | 8 (0.4)                         |
| Ethnicity, No. (%)                                  |                                 |                                 |                                 |
| Hispanic or Latino                                  | 283 (14.1)                      | 275 (13.7)                      | 392 (19.7)                      |
| Not Hispanic or Latino                              | 1688 (83.9)                     | 1689 (84.2)                     | 1576 (79.1)                     |
| Unknown or not reported                             | 40 (2.0)                        | 42 (2.1)                        | 25 (1.3)                        |
| BMI                                                 |                                 |                                 |                                 |
| Mean (SD)                                           | 30.3 (6.2)                      | 30.1 (6.1)                      | 31.1 (7.1)                      |
| Median (IQR)                                        | 29.4 (25.9-33.8)                | 29.2 (25.7-33.6)                | 30.2 (26.2-34.9)                |
| BMI group, No. (%) <sup>d</sup>                     | n = 1995                        | n = 1984                        | n = 1963                        |
| <30                                                 | 1077 (54.0)                     | 1096 (55.2)                     | 951 (48.4)                      |
| ≥30                                                 | 918 (46.0)                      | 888 (44.8)                      | 1012 (51.6)                     |
| Comorbidity group, No. (%)                          |                                 |                                 |                                 |
| High risk <sup>e</sup>                              | 1312 (65.2)                     | 1292 (64.4)                     | 1233 (61.9)                     |
| Low risk                                            | 699 (34.8)                      | 714 (35.6)                      | 740 (37.0)                      |
| Influenza vaccine received since Sept 2022, No. (%) | 1019 (50.7)                     | 1016 (50.6)                     | 783 (39.3)                      |
| COVID-19 vaccine received since Sept 2022, No. (%)  | 853 (42.4)                      | 854 (42.6)                      | 632 (31.7)                      |

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); HD-ILIV4, high-dose quadrivalent inactivated influenza vaccine; SD-ILIV4, standard-dose quadrivalent inactivated influenza vaccine.

<sup>a</sup> The safety population included all participants who were randomized and received the study vaccination. Participants were included in the vaccine group corresponding to the vaccine they received.

<sup>b</sup> Due to misrandomization, 2 participants aged 50-64 years were enrolled in the ≥65 years cohort and 3 participants aged ≥65 years were enrolled in the 50-64 years cohort.

<sup>c</sup> Race and ethnicity of participants was self-reported according to multiple categories. The total number of participants with a reported race (excluding participants with race unknown or unreported); percentages are based on this total.

<sup>d</sup> The total number of participants with a reported BMI group (excluding participants with unknown BMI group); percentages are based on this total.

<sup>e</sup> High-risk comorbidity included having any medical history of autoimmune/immune-mediated disease, blood disorders, cardiac disorders, nervous system disorders, diabetes mellitus, kidney disorders, hepatic disorders, mental impairment disorders, pulmonary disorders, or metabolism and nutritional disorders. Comorbidities based on self-report or review of medical records.

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## Tips from the consulting room



- Have your research question in mind, as well as a clear idea of what your publication goals are. Think about your ‘story’.



- If you’re planning to publish your work, be sure to review the journal’s “Instructions for authors” and browse recent articles to guide your layout and presentation!



- Consider who will be reading your work and present your findings in a way that’s easy for them to understand.
  - Is it clinicians? Is it industry? Is it other researchers? Is it a broad audience/lay people?
  - Try and make things as readable and accessible as possible.



---

# How to present your results in an accessible way

- Ensure that each figure or table you include can be understood independently, without needing to refer to the main text.
- Don't overload the reader – remember any additional information can be provided in your Appendix/Supplementary Materials and there will be maximum word counts for most journals.
- Group and order your information logically.
- Be consistent in your terminology throughout the manuscript and always provide the units of analysis. E.g. always call it “Intervention group” not “Intervention group” in one part and “Treatment group” in another, or “numerical predictor” in one part and “continuous predictor” in another.
- Consider colourblind friendly colour palettes and ensure high contrast between graph elements (e.g. lines, points).

# Best-practice reporting guidelines



- [The Equator Network](#) (Enhancing the QUALity and Transparency Of health Research) is a centralised platform for researchers to access a wide range of reporting guidelines.
  - Seeks to improve the reliability and value of published health research by promoting transparent, standardised and accurate reporting.
- There are guidelines for different study types to help you with your manuscript writing.
  - Provides you with a check-list of information required so that your manuscript can be easily understood by the reader and reproduced by other researchers.



## Reporting guidelines for main study types

|                                               |                         |                            |
|-----------------------------------------------|-------------------------|----------------------------|
| <a href="#">Randomised trials</a>             | <a href="#">CONSORT</a> | <a href="#">Extensions</a> |
| <a href="#">Observational studies</a>         | <a href="#">STROBE</a>  | <a href="#">Extensions</a> |
| <a href="#">Systematic reviews</a>            | <a href="#">PRISMA</a>  | <a href="#">Extensions</a> |
| <a href="#">Study protocols</a>               | <a href="#">SPIRIT</a>  | <a href="#">PRISMA-P</a>   |
| <a href="#">Diagnostic/prognostic studies</a> | <a href="#">STARD</a>   | <a href="#">TRIPOD</a>     |
| <a href="#">Case reports</a>                  | <a href="#">CARE</a>    | <a href="#">Extensions</a> |
| <a href="#">Clinical practice guidelines</a>  | <a href="#">AGREE</a>   | <a href="#">RIGHT</a>      |
| <a href="#">Qualitative research</a>          | <a href="#">SRQR</a>    | <a href="#">COREQ</a>      |
| <a href="#">Animal pre-clinical studies</a>   | <a href="#">ARRIVE</a>  |                            |
| <a href="#">Quality improvement studies</a>   | <a href="#">SQUIRE</a>  | <a href="#">Extensions</a> |
| <a href="#">Economic evaluations</a>          | <a href="#">CHEERS</a>  | <a href="#">Extensions</a> |



# Best-practice reporting guidelines



## Key Reasons for Adopting Reporting Guidelines

|  |                                                                                                                                                                                                                                                                 |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>STANDARDIZED REPORTING</b><br>Reporting guidelines provide a <b>standardized</b> framework for <b>documenting</b> and <b>reporting</b> experimental procedures, methods, and results. They ensure <b>consistent</b> capture of essential information.        |
|  | <b>ENHANCING REPRODUCIBILITY</b><br>Complete and transparent reporting enables other researchers to <b>replicate</b> and <b>verify</b> study findings, <b>minimizing ambiguity</b> and <b>increasing reproducibility</b> of results.                            |
|  | <b>TRANSPARENCY AND TRUSTWORTHINESS</b><br>Transparent reporting instills <b>confidence</b> , allowing readers to <b>critically evaluate</b> research methodology and <b>identify limitations, biases, or sources of error</b> .                                |
|  | <b>FACILITATING DATA SHARING AND REUSE</b><br>Reporting guidelines promote data sharing and facilitate the <b>integration</b> of existing research, accelerating <b>scientific progress</b> and fostering <b>collaboration</b> .                                |
|  | <b>IMPROVING DATA INTERPRETATION</b><br>Detailed reporting helps readers <b>accurately interpret</b> and understand presented data by providing <b>essential contextual information</b> and <b>avoiding misinterpretation</b> .                                 |
|  | <b>CONSISTENCY AND COMPARABILITY</b><br>Reporting guidelines ensure <b>consistency</b> and <b>comparability</b> within a field, aligning experiments with <b>accepted practices</b> and <b>facilitating meta-analyses and literature reviews</b> .              |
|  | <b>IDENTIFICATION OF METHODOLOGICAL BIASES</b><br>Transparent reporting enables <b>identification of biases or methodological flaws</b> that may impact <b>reliability</b> and <b>validity</b> , aiding critical assessment of the study's <b>limitations</b> . |
|  | <b>COMPLIANCE WITH ETHICAL AND REGULATORY STANDARDS</b><br>Reporting guidelines incorporate <b>ethical and regulatory considerations</b> , ensuring adherence to <b>legal, ethical, and safety obligations</b> in biomedical experimentation.                   |

Best practices for data management and sharing in experimental biomedical research



# Statistical analysis plans (SAPs)



- SAPs as well as study protocols are a great tool to improve communication and collaboration between you and other researchers.
- They should include information on:
  - Objectives and hypotheses
  - Primary and secondary outcomes
  - Study design
  - Data collection methods
  - Data management
  - Statistical analysis including handling of missing data and sensitivity analyses
  - Reporting of findings
- [Australian Clinical Trials Alliance: Statistical analysis plan](#)
- [Jama: Guidelines for statistical analysis plans](#)
- [A template for the authoring of statistical analysis plans](#)
- [Developing a Quantitative Data Analysis Plan for Observational Studies](#)

---

# A cautionary tale...

*When inexperience meets bad statistical advice*

**This is based on a true story from the consulting room highlighting the importance of Research Essentials...**

*A client approached me for a stats consult with a manuscript due in one week. The whole paper was written, and they just needed confirmation of one results section. The client had relied on a collaborator to analyse the data (who had now left), and they were also under pressure to submit, trusting that the work completed was sound. Whilst sharing their screen and reading through their manuscript and r code, alarm bells started ringing in my head. There were a variety of statistical issues, highlighting a lack of statistical literacy, reproducibility and academic integrity from the collaborator.*



---

# A cautionary tale...

*When inexperience meets bad statistical advice*

**Data import & cleaning:** Raw REDCap .csv imported directly with no data cleaning prior to analysis.

**Outlier handling:** Plausible 'outlier' minimum values were removed before analysis, without justification or sensitivity analyses.

**Variable coding:** Variable types not defined so all predictors were treated as numeric. Categories did not match data dictionary, making interpretation unclear.

**Exploratory data analysis:** Limited EDA, histograms plotted, but for categorical variables coded as numeric.

**Statistical terminology:** "Multivariate" incorrectly used instead of "multivariable".

**Model specification:** Models violated assumptions, had sparse cell counts, and were overly complex and not parsimonious.

**Reproducibility:** Participant data altered or omitted, and summary tables could not be replicated.

**Data management:** Files were missing or poorly named in the data folders and multiple result versions were shared with the client.

**P-value focus:** Heavy reliance on arbitrary p-value thresholds, with little attention to effect size or clinical relevance.

**Final advice:** Client was advised to include a disclaimer that results should be interpreted with caution as they may be unreliable.

---

# A cautionary tale...

## **Due diligence matters!**

- Even if you are not the primary one analysing the data, if your name is on the paper, then you are also responsible for the academic integrity of the work.
- Many journals now require author contribution statements, making clear who did what, but nonetheless, shared accountability still applies.
- Basic statistical checks (e.g. data types, summary tables and EDA) are often enough to uncover major issues early on and will save you time down the track.
- Don't overlook the simple stuff, as it can catch what complex analysis might obscure.
- *Start simple → get complex.*

---

## Some analysis examples

**5. Exploratory data analysis (EDA)**

**6. Inferential analysis**

---

## Data analysis workflow: 4 examples

- **A.** Linear models examples – Simple regression, ANOVA, ANCOVA, Repeated measures.
- **B.** Extended linear models example – Survival analysis.
- **C.** Extended linear models example – Generalised linear model (Binary logistic regression)
- **D.** Multivariate analysis – Confirmatory factor analysis (CFA)

---

# Your variable types will dictate the type of statistical analysis you perform

- **Examples using the generalised linear model framework:**
  - 2-sample t-test = binary explanatory variable and numerical outcome.
  - ANOVA = multi-categorical explanatory variable and numerical outcome.
  - Simple linear regression = numerical explanatory variable and numerical outcome.
  - Binary logistic regression = numerical or categorical explanatory variable and binary outcome.
- For more information on the type of statistical test to run, see slide 98!

---

## Example A: Linear models examples

**Scenario: We are interested in studying a numerical (continuous) outcome variable, e.g. weight gain (kg) or blood cell count (cells/ $\mu$ L).**

- 1. Simple linear regression – one numerical predictor variable.
- 2. ANOVA (control vs. treatment) – for 2 groups = 2 sample t-test = simple linear regression with one binary predictor variable.
- 3. ANCOVA – ANOVA with a covariate predictor.

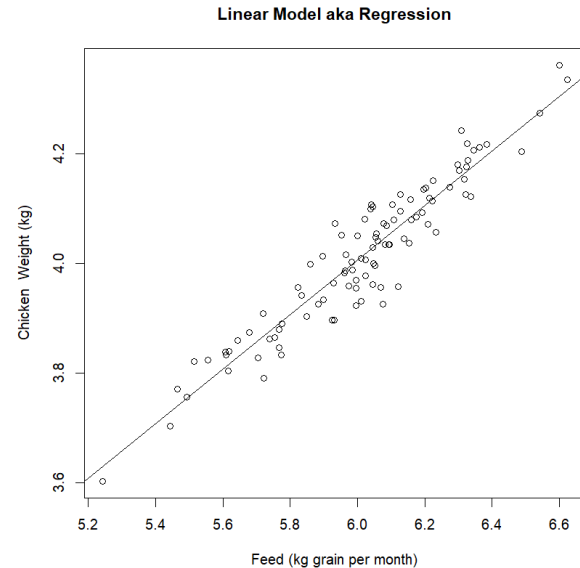
For more detail on how to do these analyses including R code, attend our SIH [Linear Models 1: Linear Regression, ANOVA, ANCOVA and Repeated Measures \(a Simple Mixed Model\)](#) workshop!

---

# Example A1: Linear models – Simple linear regression

**Step 5: EDA – Plot the data in a scatterplot.**

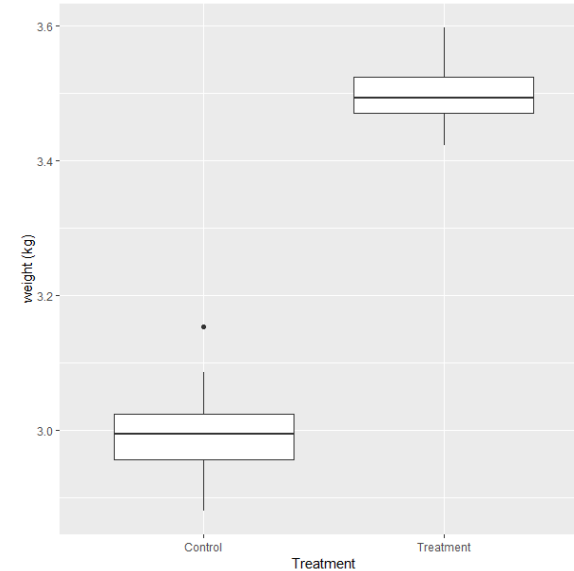
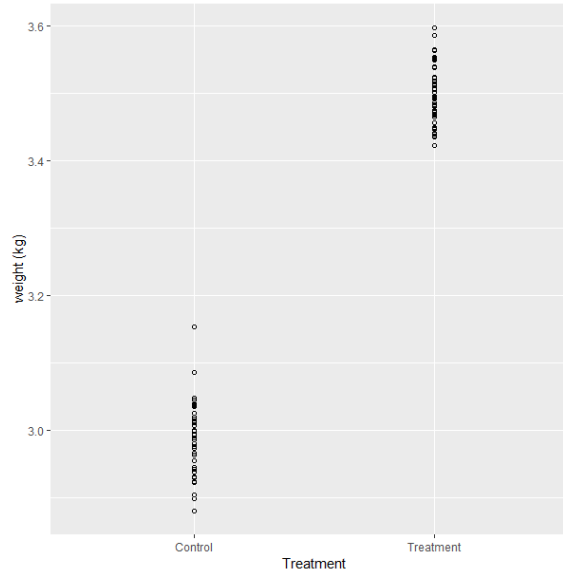
**Step 6: Inferential analysis – fit a linear regression line and test if the slope is different from 0;  $P < 0.001$ ; report slope/regression estimate and 95% CI.**



## Example A2: Linear models – Control vs. treatment experiment

**Step 5: EDA – Plot the data with grouped boxplots.**

**Step 6: Inferential analysis – ANOVA/2-sample t-test;  $P < 0.001$ ; report predicted means and 95% CIs.**

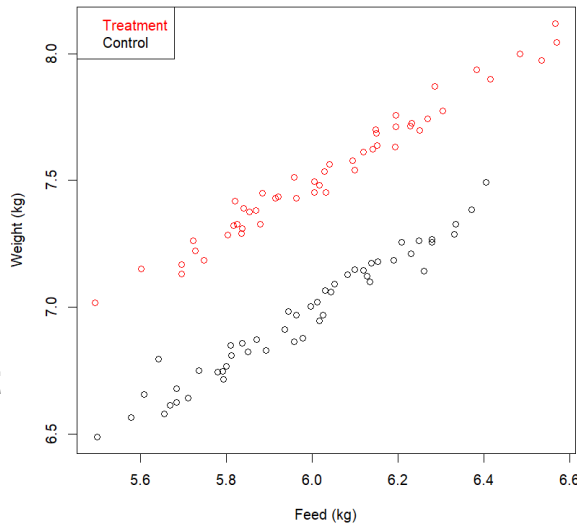


## Example A3: Linear models – ANCOVA (ANOVA with a numerical covariate)

**Step 5: EDA – Plot the data**  
(differentiate categories of the treatment variable).

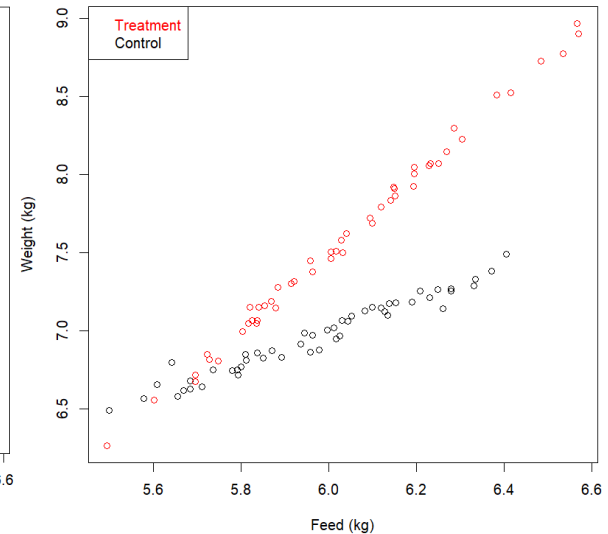
**Step 6: Inferential analysis – ANCOVA or multivariable regression;  $P < 0.001$ ; report predicted means with or without interaction and 95% CIs.**

Supplement has no impact on feed's relationship with weight



Without interaction

Supplement has an impact on feed's relationship with weight



With interaction

---

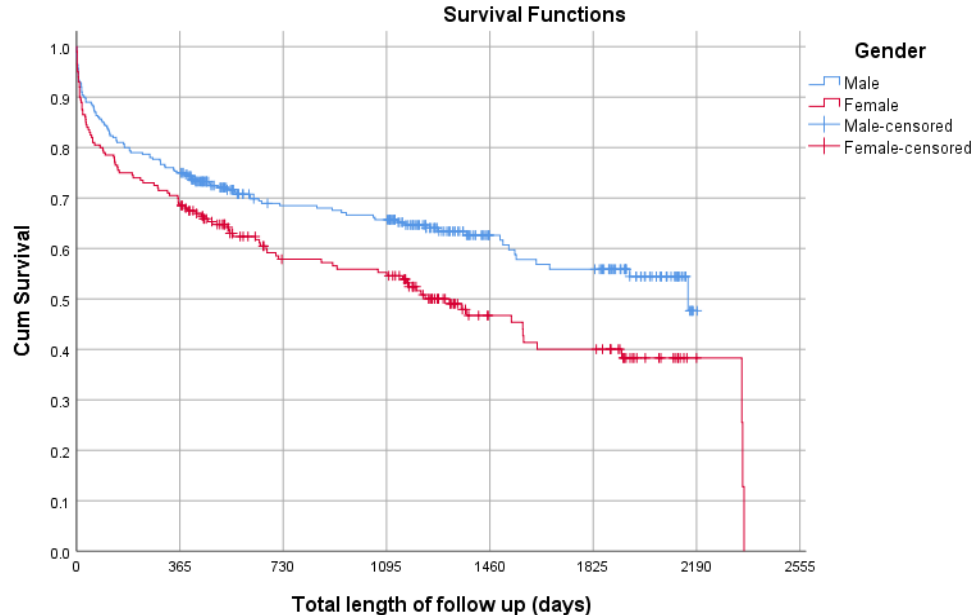
## Example B: Survival analysis

- **Scenario:** Worcester Heart Attack Community Surveillance Study (WHAS)
- **Aim:** To examine time trends in the incidence rate of acute heart attacks.
- **Objective:** Investigate if different demographic and clinical factors are associated with the time to a heart attack.
- **Data:** Longitudinal, observational data.
- **Outcome:** Heart attack – time to event.
- **Predictors:** Demographic and clinical data.
- **Key feature:** Data is censored – see our Introduction to Survival Analysis WS.

---

## Example B: Survival analysis

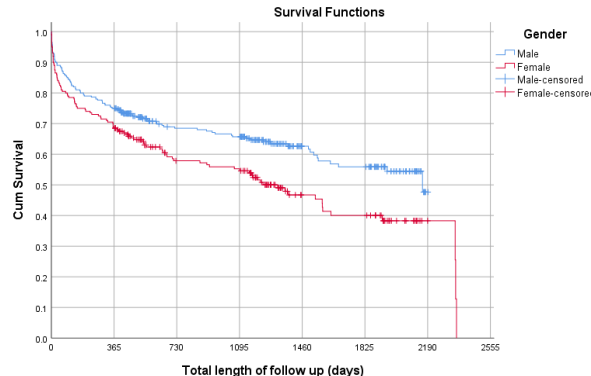
**Step 5: EDA – Kaplan Meier curve is the EDA plot for survival analysis.**



## Example B: Survival analysis

Step 6: Inferential analysis – The logrank test to compare the survival curves of two or more groups.

- There is a significant difference in survival between males and females (by log-rank test).
  - Median survival for males: 2160 days [95%CI: not calc].
  - Median survival for females: 1317 days [95% CI 970-1664].



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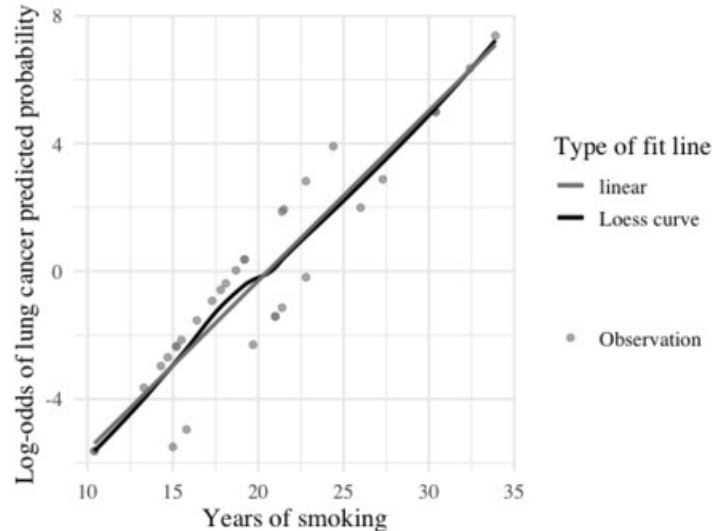
## Example C: Binary logistic regression

- **Scenario:** Survey responses of 32 long-term smokers to determine the association between smoking status and lung-cancer diagnosis.
- **Outcome:** Lung cancer (no/yes)
- **Predictors:** Years smoking, BMI
- [Primer on binary logistic regression](#)

For more detail on how to do these analyses including R code, attend our SIH [Linear Models 2: Logistic and Poisson/Count Regression - An Introduction to Generalised Linear Models](#) workshop!

## Example C: Binary logistic regression

**Step 5: EDA – Log odds of the outcome (lung cancer) vs. numerical predictor (years of smoking) to check the linearity assumption.**



**Figure 2**

Checking the linearity assumption graphically.

---

## Example C: Binary logistic regression

**Step 6: Inferential analysis – binary logistic regression to examine association between lung cancer (y/n) and years smoking (numerical), if linearity assumption is met.**

For every additional year of smoking, the odds of lung cancer are approximately 1.98 times higher (95% CI 1.36 to 3.86).

---

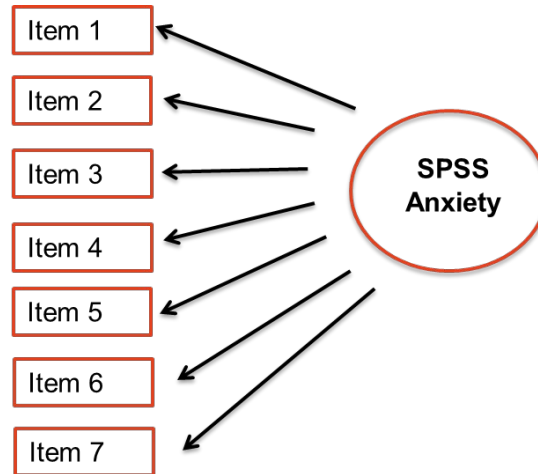
## Example D: Multivariate analysis – Confirmatory factor analysis

- **Scenario:** To test if a factor model 'SPSS statistical software Anxiety' explains the common variance among 7 questionnaire items:
  1. I dream that Pearson is attacking me with correlation coefficients.
  2. I have little experience with computers.
  3. All computers hate me.
  4. I have never been good at mathematics.
  5. My friends are better at statistics than me.
  6. Computers are useful only for playing games.
  7. I did badly at mathematics at school.

---

## Example D: Multivariate analysis – Confirmatory factor analysis

- **Scenario:** Confirm SPSS Anxiety as a factor explaining the common variance among the 7 items.



Example adapted from [A Practical Introduction to Factor Analysis: Confirmatory Factor Analysis](#) from the UCLA: Statistical Consulting Group.

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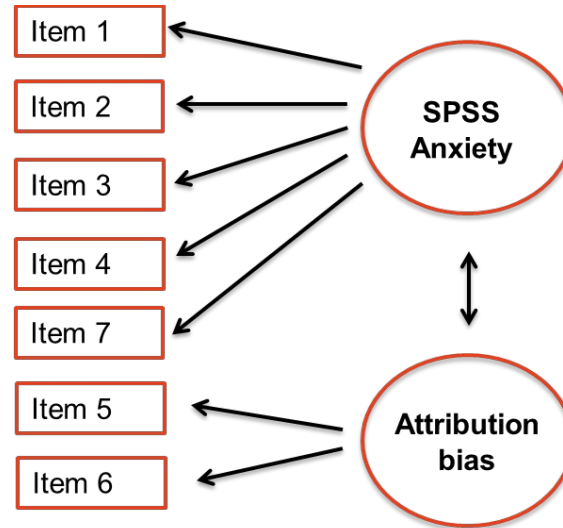
## Example D: Multivariate analysis – Confirmatory factor analysis

**Step 5:** EDA – scatter plots + Pearson's correlation coefficient  $r$ ; correlation matrix.

|    | Q1    | Q2    | Q3   | Q4   | Q5          | Q6   | Q7 |
|----|-------|-------|------|------|-------------|------|----|
| Q1 | 1     |       |      |      |             |      |    |
| Q2 | -0.34 | 1     |      |      |             |      |    |
| Q3 | 0.44  | -0.38 | 1    |      |             |      |    |
| Q4 | 0.40  | -0.31 | 0.40 | 1    |             |      |    |
| Q5 | 0.22  | -0.23 | 0.28 | 0.26 | 1           |      |    |
| Q6 | 0.31  | -0.38 | 0.41 | 0.34 | <b>0.51</b> | 1    |    |
| Q7 | 0.33  | -0.26 | 0.35 | 0.27 | 0.22        | 0.30 | 1  |

## Example D: Multivariate analysis – Confirmatory factor analysis

**Step 6:** Inferential analysis – to test if a 2-factor model ‘SPSS statistical software Anxiety’ and ‘Attribution bias’ explains the common variance among 7 questionnaire items.





## Final notes on Step 6: Inferential analysis

- **We only showed some common examples of statistical analyses – there are many different types of analyses you can consider:**
  - Other Linear Models extensions such as Poisson regression and more complex mixed models - see our SIH [Linear Models 2: Logistic and Poisson/Count Regression - An Introduction to Generalised Linear Models](#) workshop.
  - Survival Analysis for ‘time-to-event’ outcome data – see our SIH [Introduction to Survival Analysis](#) training!
  - Survey Data analysis – see our [Design and Analysis of Surveys 1](#) and [Design and Analysis of Surveys 2: Advanced Topics](#).
  - Other Multivariate Analyses – for example PCA, Factor Analysis - see our [Multivariate Statistical Analysis 1: Dimension Reduction](#).
  - And more [more workshops](#)!



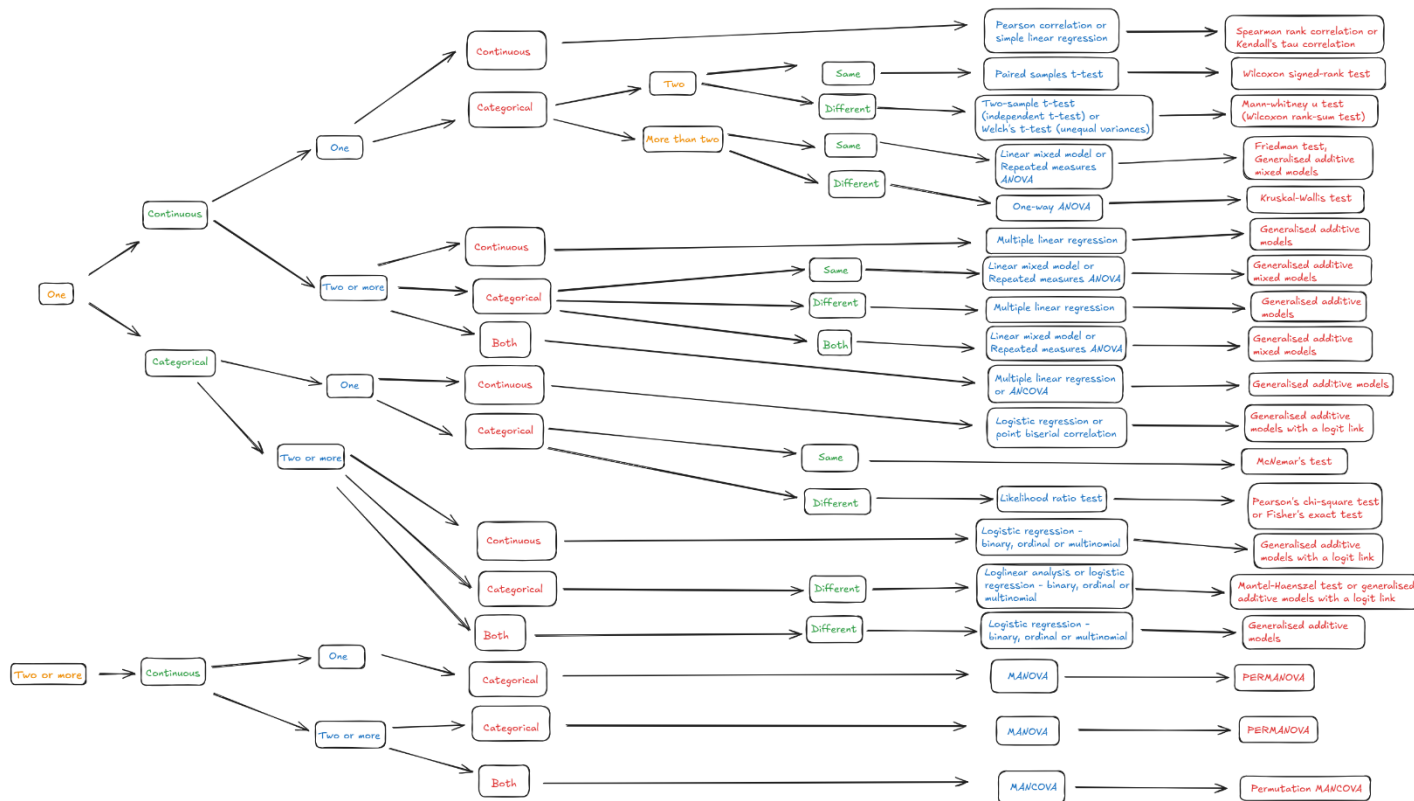
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## Final notes on Step 6: Inferential analysis

- Start simple and increase complexity step-by-step.
- Always consider/check the test/model assumptions.
- Report 95% CIs for estimates, e.g., predicted means/probabilities/rates.
- For basic analyses consider more powerful (parametric) analyses first and use less powerful tests if assumptions are violated. e.g.:
  - Use a 2-sample t-test with equal or unequal variance for means, before a Mann-Whitney test.
  - Use a chi-squared test to compare proportions before a Fisher's exact test.
- Use knowledge of variable types to guide you through the systematic roadmap on the next slide!



Assumptions of  
linear model partially  
or fully unmet?  
Use semiparametric  
or nonparametric.



Where possible, if the assumptions are met, use the parametric test as it has more statistical power to detect differences than the nonparametric test equivalent.

\*If you are unsure of the units in your study, see our [Experimental Design](#) workshop for more information!



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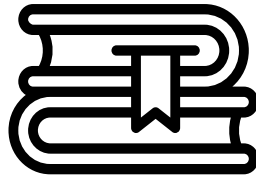
## Final notes on Step 6: Inferential analysis

- **Univariate/univariable**  
– involving one variable, e.g. one outcome per analysis; analysis with one predictor variable.
- **Multivariate** – multiple outcomes in the same analysis/model.
- **Multivariable** – multiple explanatory variables in the same analysis/model.
- **Linear models** (LM – numerical outcome).
- **Generalised linear models** (GLM – categorical outcomes, e.g. binary, ordinal, multinomial (for nominal outcome data) or Poisson/negative binomial regression (for count/rate outcome data)).
- **Mixed models** (i.e. LM or GLM with a random effect = LMM or GLMM) where data are clustered in space or time, e.g. repeated measures/longitudinal data.

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## Further R resources

- There is a large online community of R users contributing to free ‘packages’ with data analysis functions, which leads to many ways of coding your analysis in R. This can be confusing. We recommend using tidyverse packages and tidy-centric code.
- See our SIH helpful links for guides on using [R and Rstudio](#).
- [LinkedIn Learning: R courses](#)
  - Including [Learning the R Tidyverse \(2024\)](#), [Complete Guide to R: Wrangling, Visualizing, and Modelling Data](#), and [Cleaning Bad Data in R](#).
- [RLadiesSydney: RYouWithMe](#)



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## Other resources

### Books on R

- [R for Data Science](#) by Hadley Wickham.

### Statistical blogs and websites

- [The Analysis Factor](#)
  - [Seven Steps for Data Cleaning](#)
  - [Best Practices for Organizing your Data Analysis](#)
  - [Best Practices for Data Preparation](#)
  - [Preparing Data for Analysis is \(more than\) Half the Battle](#)
  - [Four Weeds of Data Analysis That are Easy to Get Lost In](#)

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# Further assistance at The University of Sydney



## SIH

- [Statistical Resources](#) website: containing our workshop slides and our favourite external resources (including links for learning R and SPSS).
- [Hacky Hour](#): an informal monthly meetup for getting help with coding or using statistics software.
- 1on1 Consults can be requested on our website or [here](#) (click on the big red 'contact us' link).

## SIH Workshops

- Create your own custom programs tailored to your research needs by attending more of our Statistical Consulting workshops. Look for the statistics workshops on our training page or on our [Training calendar](#).
- Sign up to our mailing list to be notified of upcoming training.

## Other

- Open Learning Environment (OLE) courses
- LinkedIn Learning

# How to engage with us



[sydney.edu.au/informatics-hub](https://sydney.edu.au/informatics-hub)

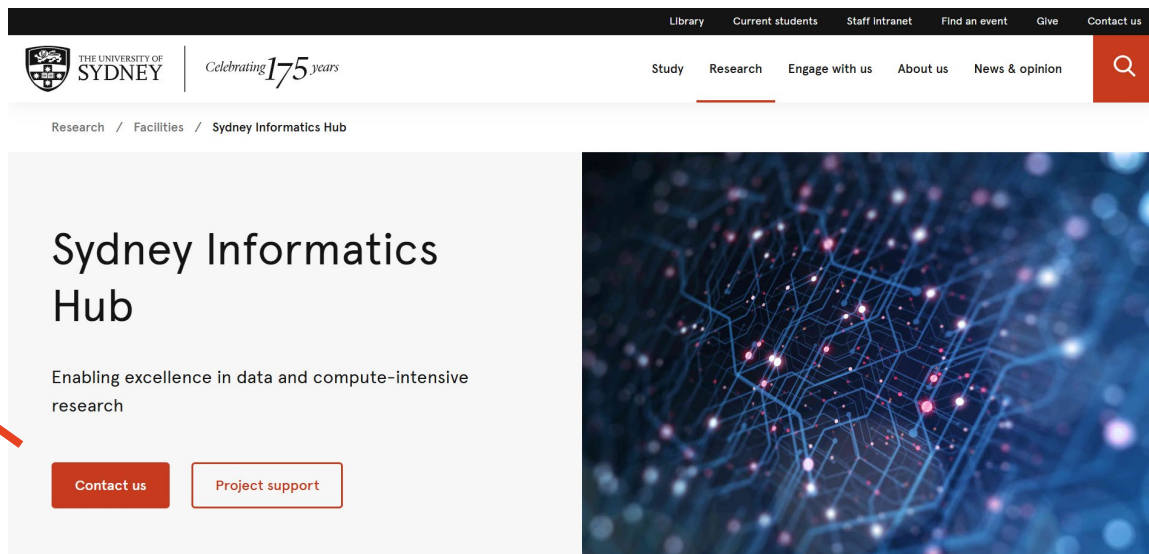


And fill out our request form



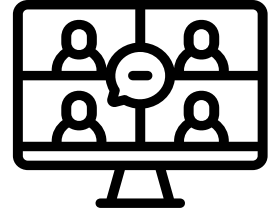
Or send us an email:  
[sih.info@sydney.edu.au](mailto:sih.info@sydney.edu.au)

The University of Sydney



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# How to use our workshops



- Workshops developed by the Statistical Consulting Team within the Sydney Informatics Hub form an integrated modular framework. Researchers are encouraged to choose modules to **create custom programs tailored to their specific needs**. This is achieved through:
  - Short 90-minute workshops, acknowledging researchers rarely have time for long multi day workshops.
  - Providing statistical workflows applicable in any software, that give practical step by step instructions which researchers return to when analysing and interpreting their data or designing their study e.g. workflows for designing studies for strong causal inference, model diagnostics, interpretation and presentation of results.
  - Each one focusing on a specific statistical method while also integrating and referencing the others to give a holistic understanding of how data can be transformed into knowledge from a statistical perspective from hypothesis generation to publication.

For other workshops that fit into this integrated framework, refer to our training link page under statistics, found below:

[Workshops and training](#)

# Online statistical consulting resources

*Sydney Informatics Hub*



[Sydney Informatics Hub: Statistical Resources](#)

## SIH Statistical Resources

[Collapse All](#) | [Expand All](#)

### Sites

☐ Welcome

 Workshops and Workflows

 The Statistical Consulting Unit

 Helpful Links

**Workshops and Workflows:** Download all our workshops with their step-by-step workflows on how to do analysis.

**References:** 1-3 Curated links on a variety of common statistical concepts and tests e.g.. Basic Theory, Meta Analysis, Software, etc. ***A great place to start looking for help!!***

**Statistical Consulting Unit/What to expect in a consult:** For tips on how to get the most out of your consult.

(You may need to click “Expand all” on the Sites sidebar to get an overview of the available pages).

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# We recommend our experimental design and sample size workshops

## Experimental design workshop

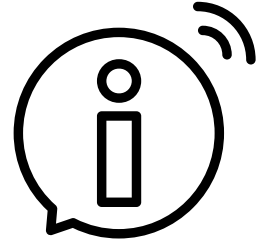
- Far too many researchers think they know all they need to in this area. We commonly see designs that could be substantially improved for stronger causal inference and improved results which leads to publication in higher impact journals (amongst other benefits).
- Even if you have already collected your data, it is well worth attending since it may improve your write up and analysis. E.g., we had a client who didn't realise they had a very strong before/after control/impact (BACI) design.

## Power and sample size workshop

- Shows the steps and decisions researchers need to make when designing an experiments to ensure sufficient sample e.g., power, minimum sample required to fit the necessary model, etc.
- Also, how much power the study has, i.e., does it have sufficient power to detect the effects you expect to see, or is your study a complete waste of time and resources?

---

## A reminder: Acknowledging SIH



- All University of Sydney resources are available to researchers free of charge.
- The use of the SIH services including the Artemis HPC and associated support and training warrants acknowledgement in any publications, conference proceedings or posters describing work facilitated by these services.
- The continued acknowledgment of the use of SIH facilities ensures the sustainability of our services.

### **Suggested wording for use of workshops and workflows:**

- *“The authors acknowledge the Statistical workshops and workflows provided by the Sydney Informatics Hub, a Core Research Facility of the University of Sydney.”*

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# We value your feedback



- We want to hear about you and whether this workshop has helped you in your research. What worked and what didn't work.
- We actively use the feedback to improve our workshops.
- Completing this survey really does help us and we would appreciate your help! It only takes a few minutes to complete (promise!)
- You will receive a link to the anonymous survey by email.